K-Means and Hierarchical Clustering

COMP 562

Junier B. Oliva

Borrowed from: http://www.cs.cmu.edu/~awm/tutorials (as well as slides from Seyoung Kim, William Cohen, Ziv Bar-Joseph, Eric Xing).

Two Classes of Learning Problems

- Supervised learning = learning from labeled data, where class labels (in classification) or output values (regression) are given
  - Train data: (X, Y) for inputs X and labels Y

- Unsupervised learning = learning from unlabeled, unannotated data
  - Train data: X for unlabeled data
  - we do not have a teacher that provides examples with their labels

Unsupervised learning

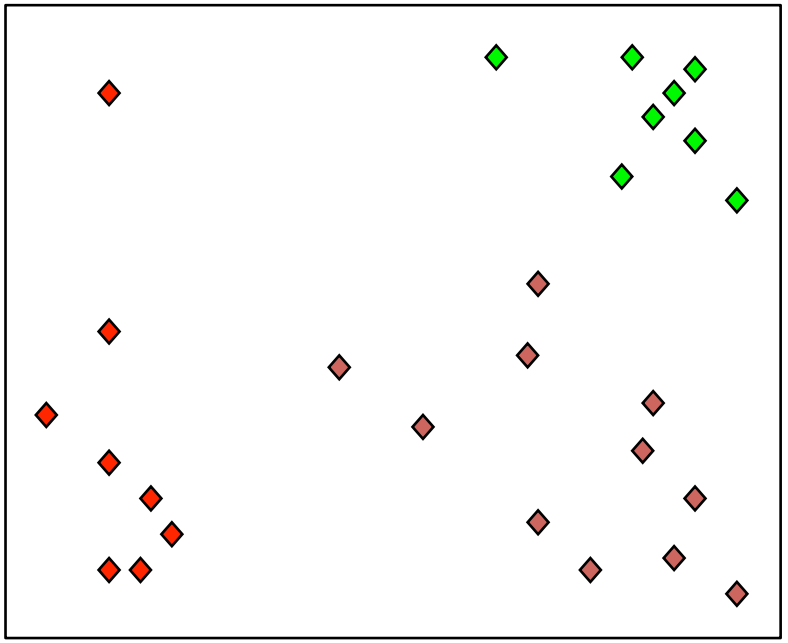
- Clustering methods
  - Non-probabilistic method
    - Hierarchical clustering
    - K means algorithm
  - Probabilistic method
    - Mixture model

- We discussed density estimation, another unsupervsed learning task

- We will also discuss dimensionality reduction, another unsupervised learning method later in the course

What is Clustering?

- Organizing data into *clusters* such that there is
  - high intra-cluster similarity
  - low inter-cluster similarity
- Informally, finding natural groupings among objects.



- Why do we want to do that?
- Any REAL application?

What does it mean to compress a d-dimensional real-valued point?

How many bits to transmit a d-dimensional point?

Lossy Compression

Suppose you transmit the coordinates of points drawn randomly from this dataset.

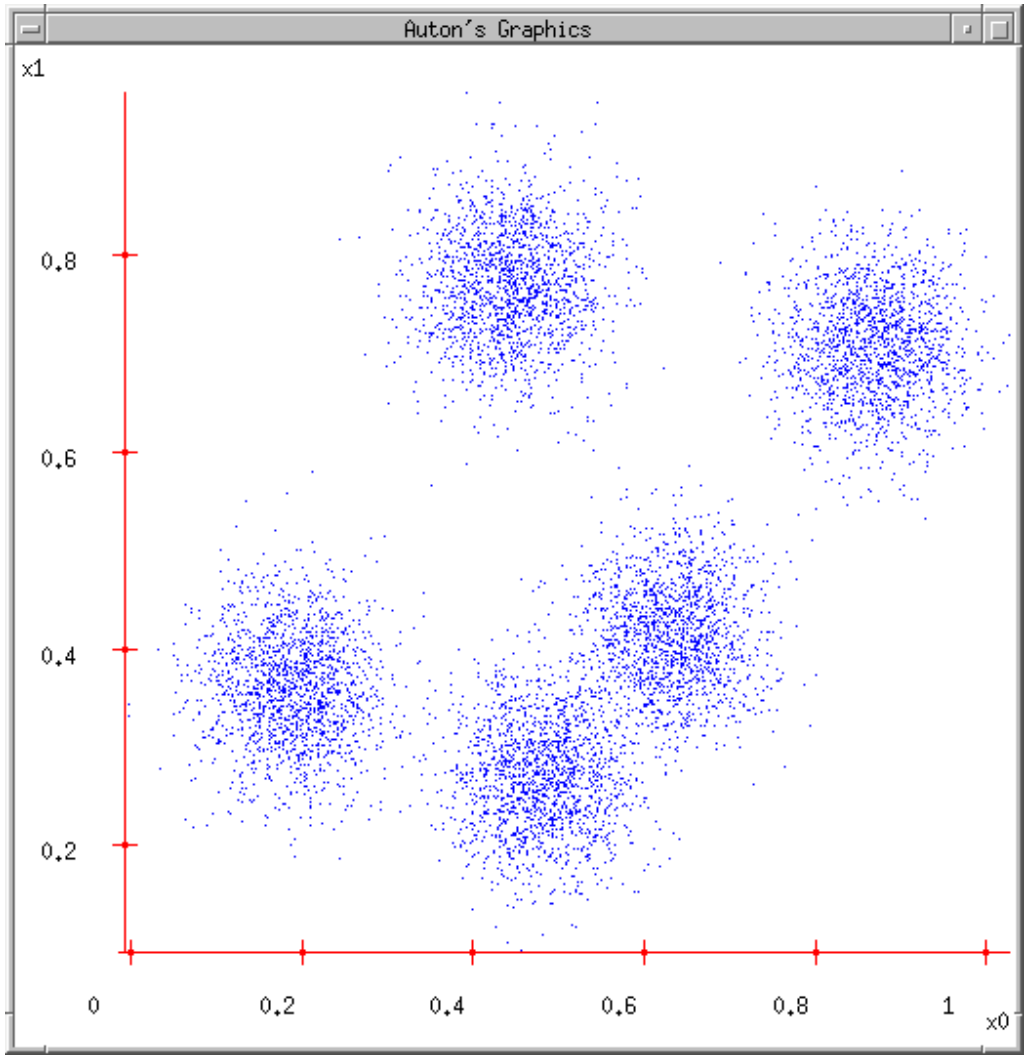
You can install decoding software at the receiver.

You're only allowed to send two bits per point.

It'll have to be a "lossy transmission".

Loss = Sum Squared Error between decoded coords and original coords.

What encoder/decoder will lose the least information?



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Idea One

Suppose you transmit the coordinates of points randomly from this d

You can install decoc software at the receiver.

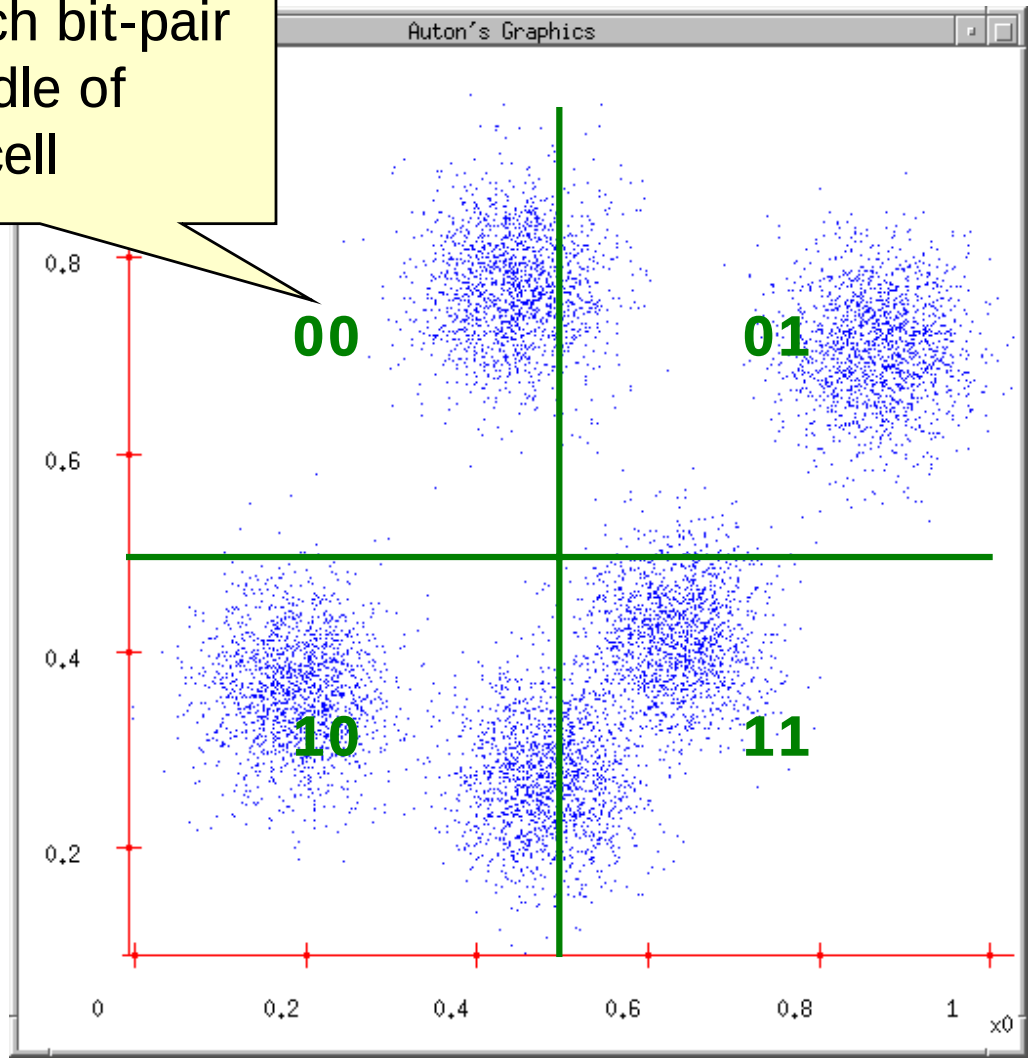
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What encoder/decoder will lose the least information?

Break into a grid, decode each bit-pair as the middle of each grid-cell



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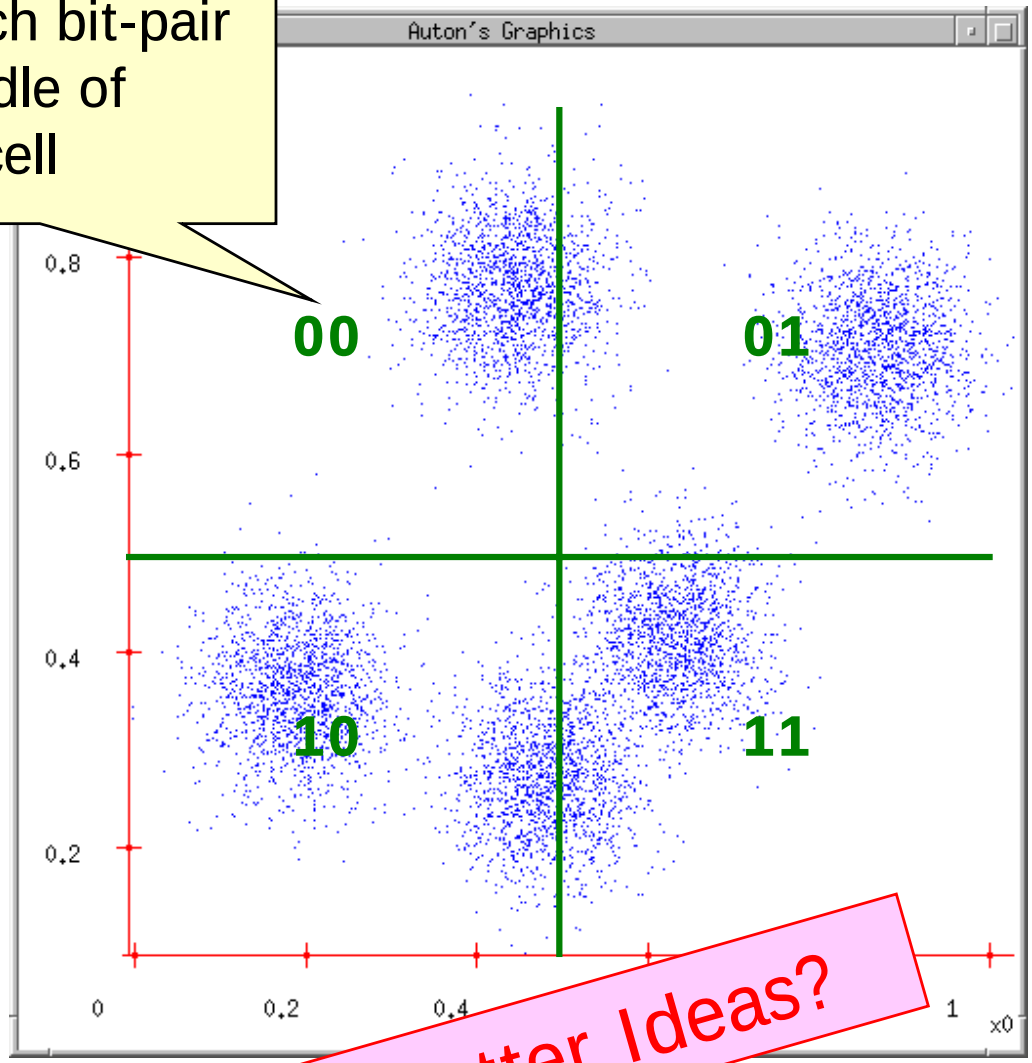
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What encoder/decoder will lose the least information?

Break into a grid, decode each bit-pair as the middle of each grid-cell



Any Better Ideas?

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Idea Two

Suppose you transmit the coordinates of poi
randomly from thi

You can install de
software at the receiver.

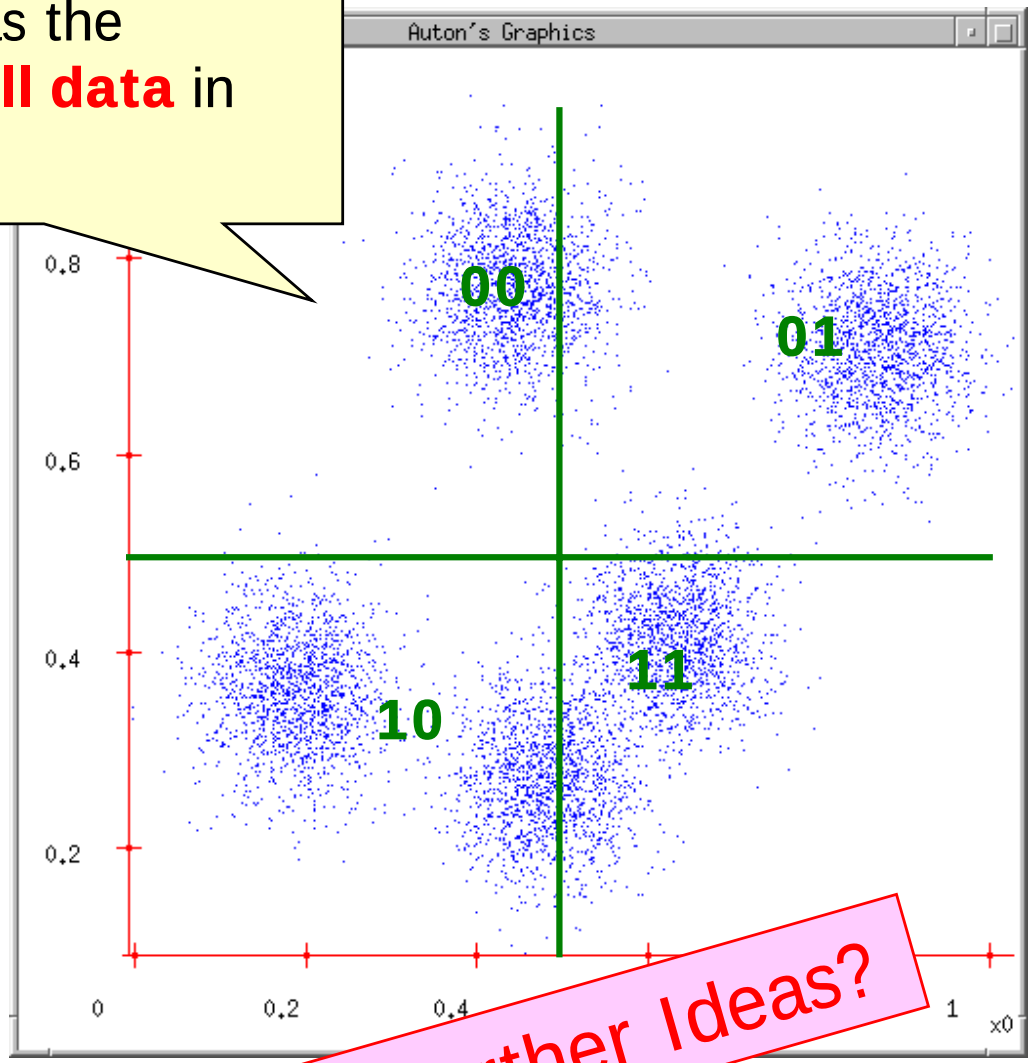
You're only allowed to send two bits per point.

It'll have to be a "lossy transmission".

Loss = Sum Squared Error between decoded coords and original coords.

What encoder/decoder will lose the least information?

Break into a grid, decode each bit-pair as the **centroid of all data** in that grid-cell

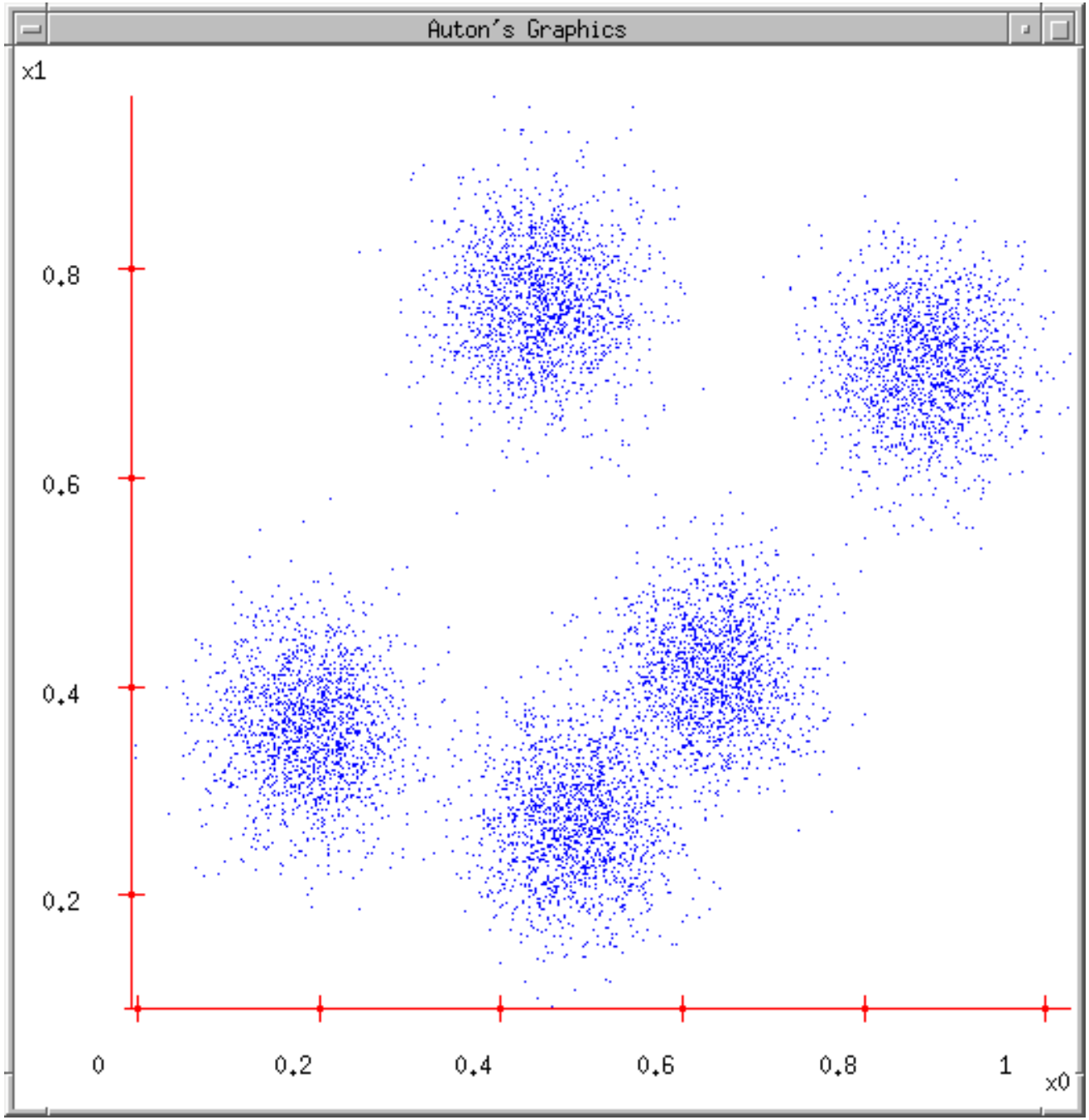


Any Further Ideas?

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K-means

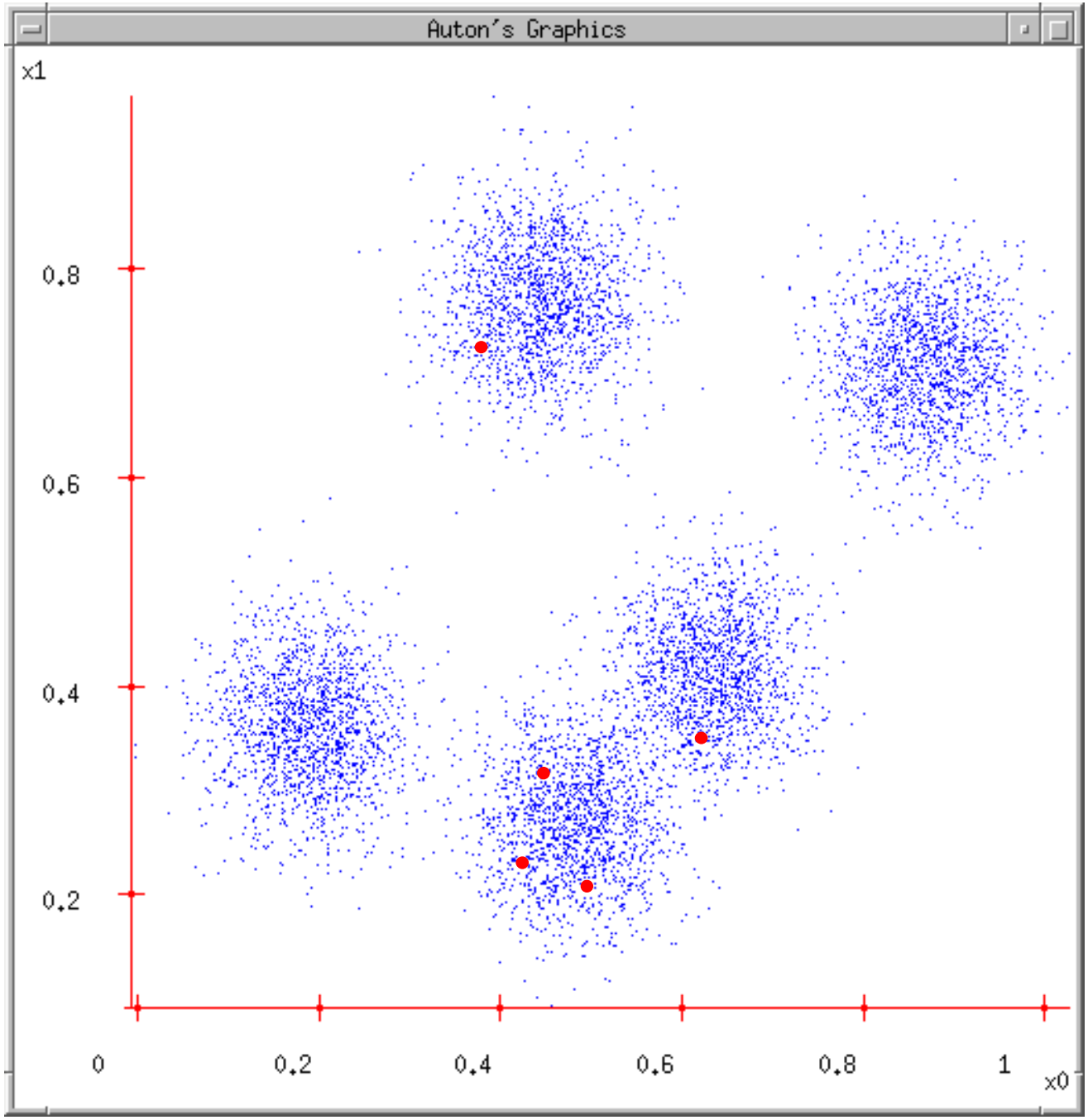
1. Ask user how many clusters they'd like. *(e.g. k=5)*



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K-means

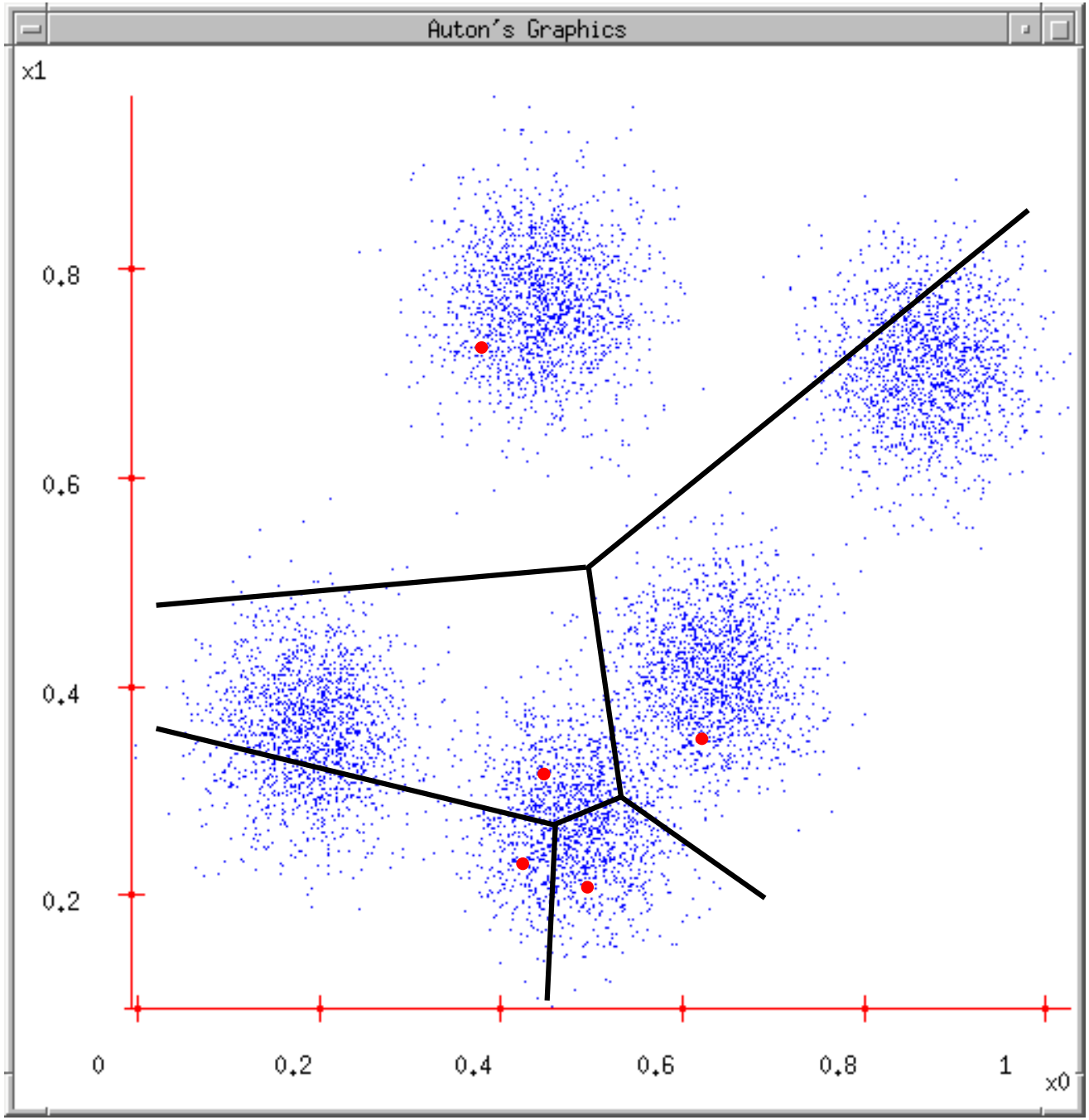
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations



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K-means

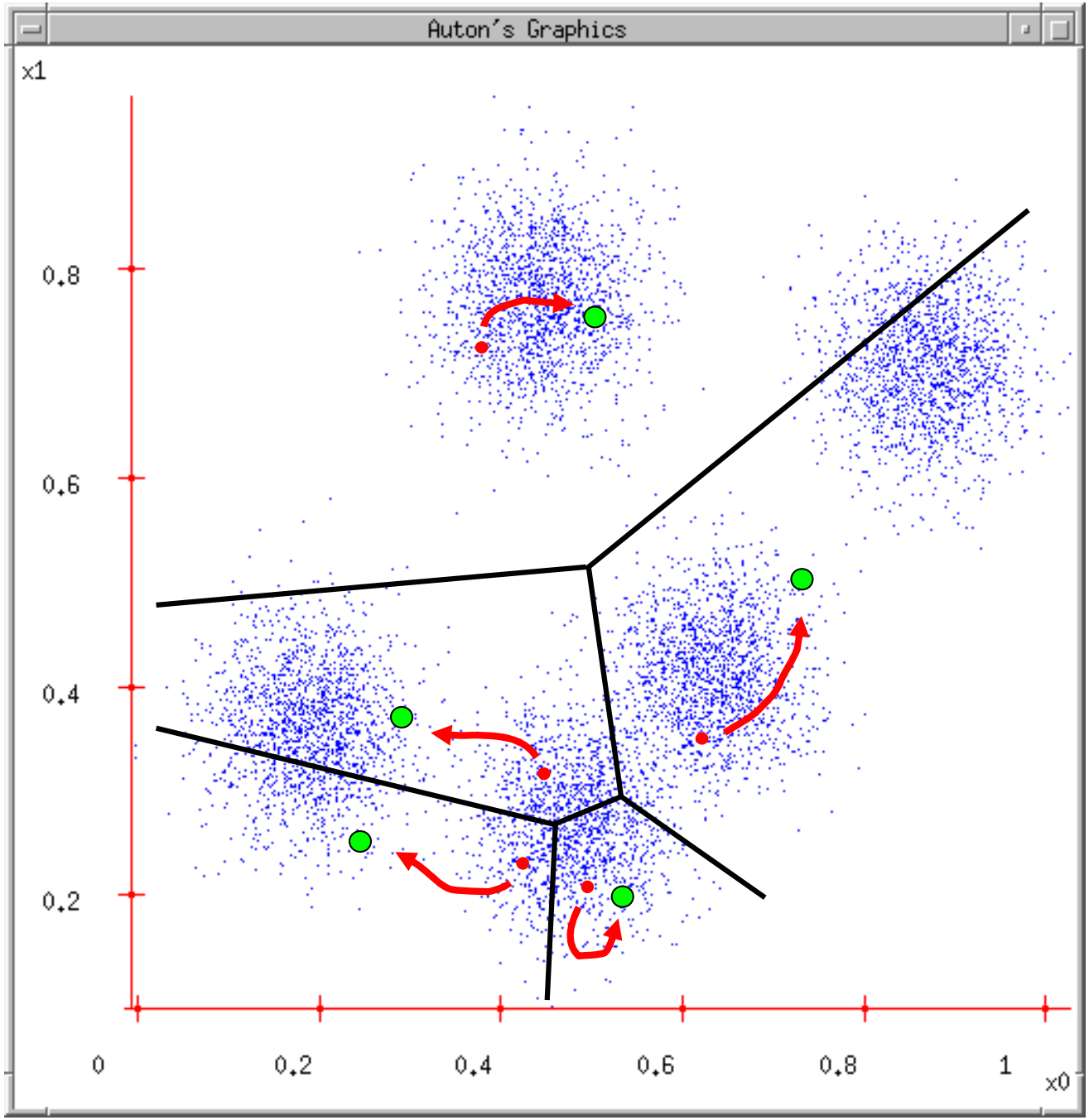
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to. (Thus each Center "owns" a set of datapoints)



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K-means

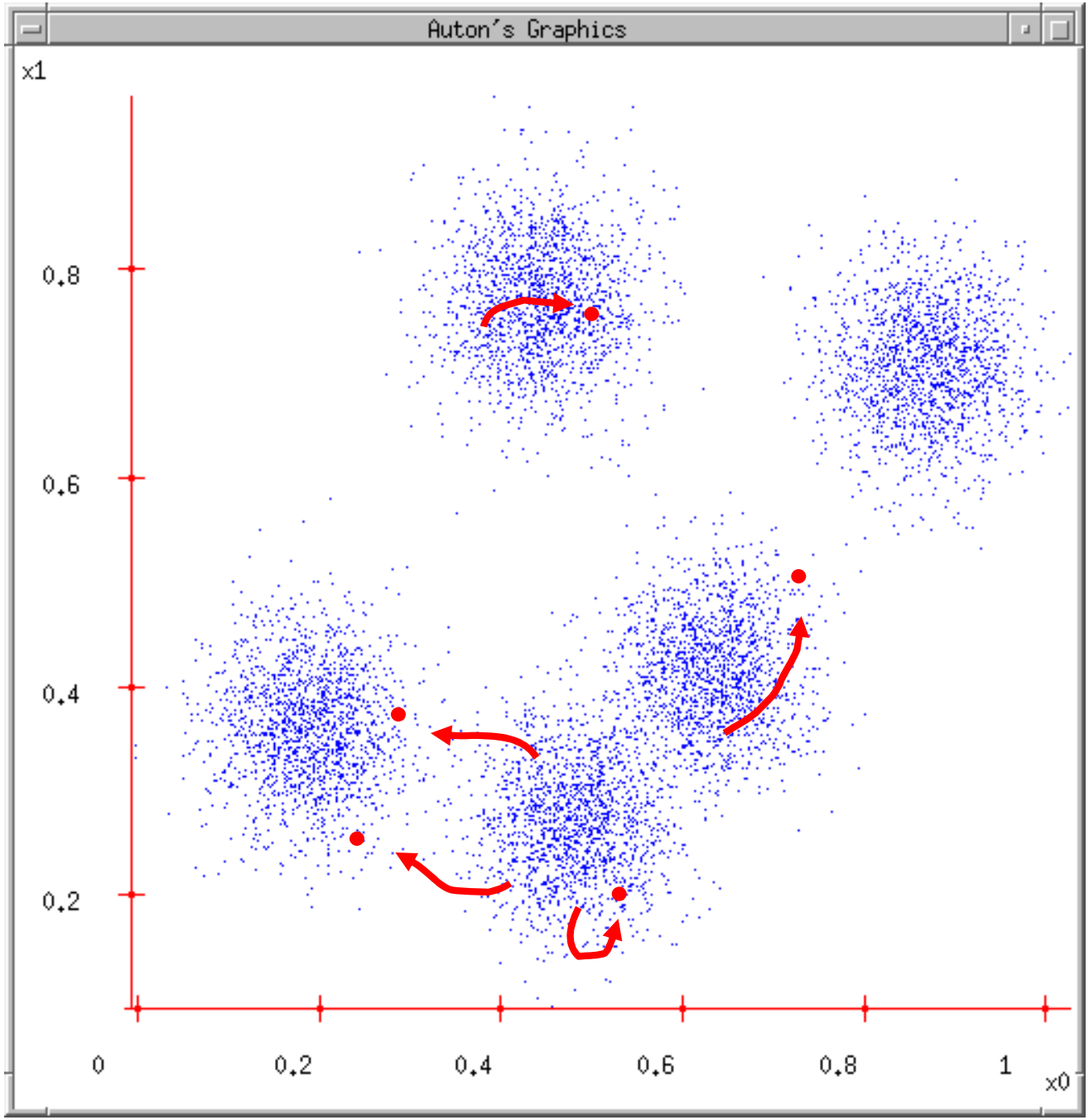
1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns



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K-means

1. Ask user how many clusters they'd like. *(e.g. k=5)*
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns...
5. ...and jumps there
6. ...Repeat until terminated!



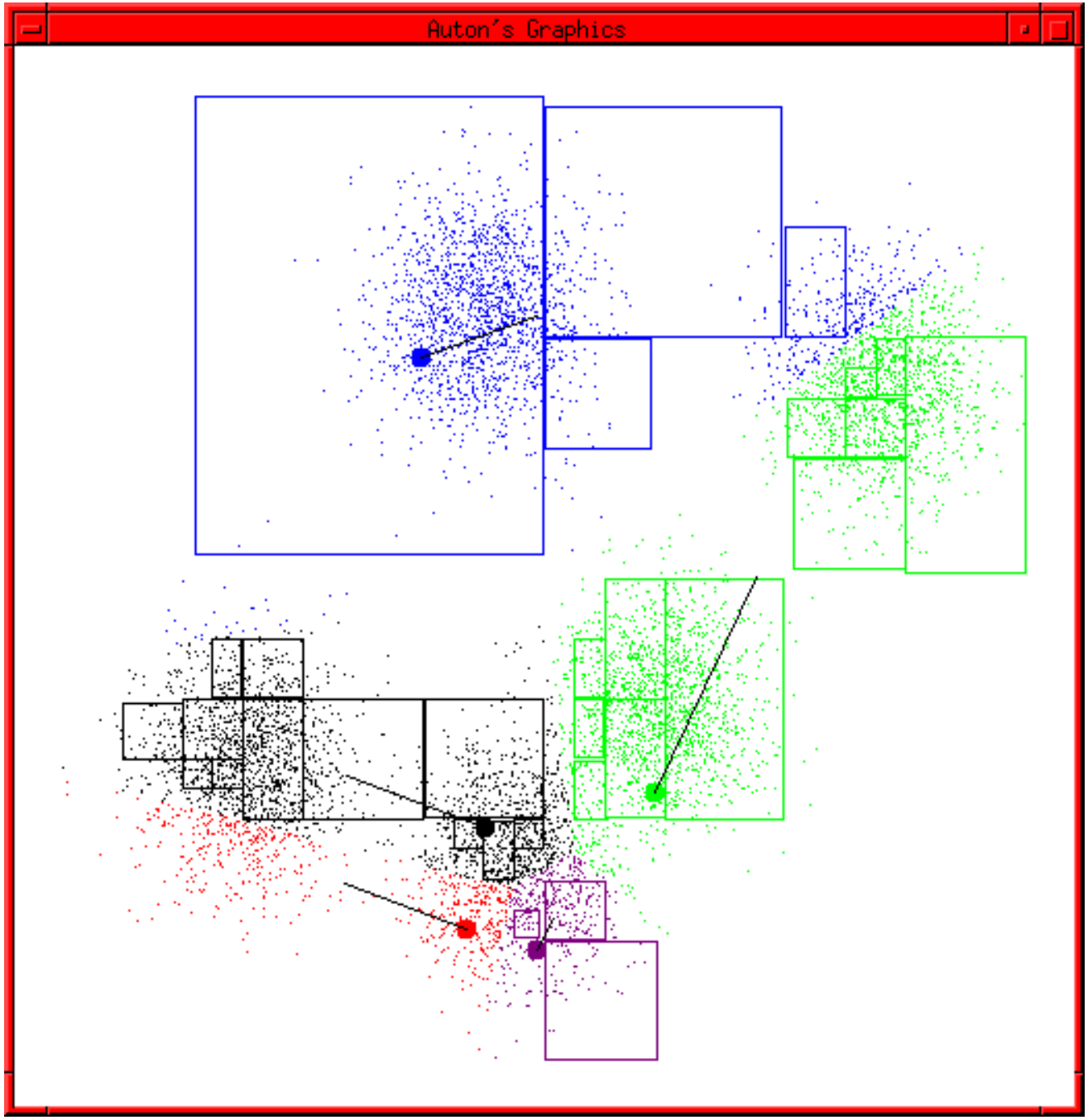
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K-means Start

Advance apologies: in Black and White this example will deteriorate

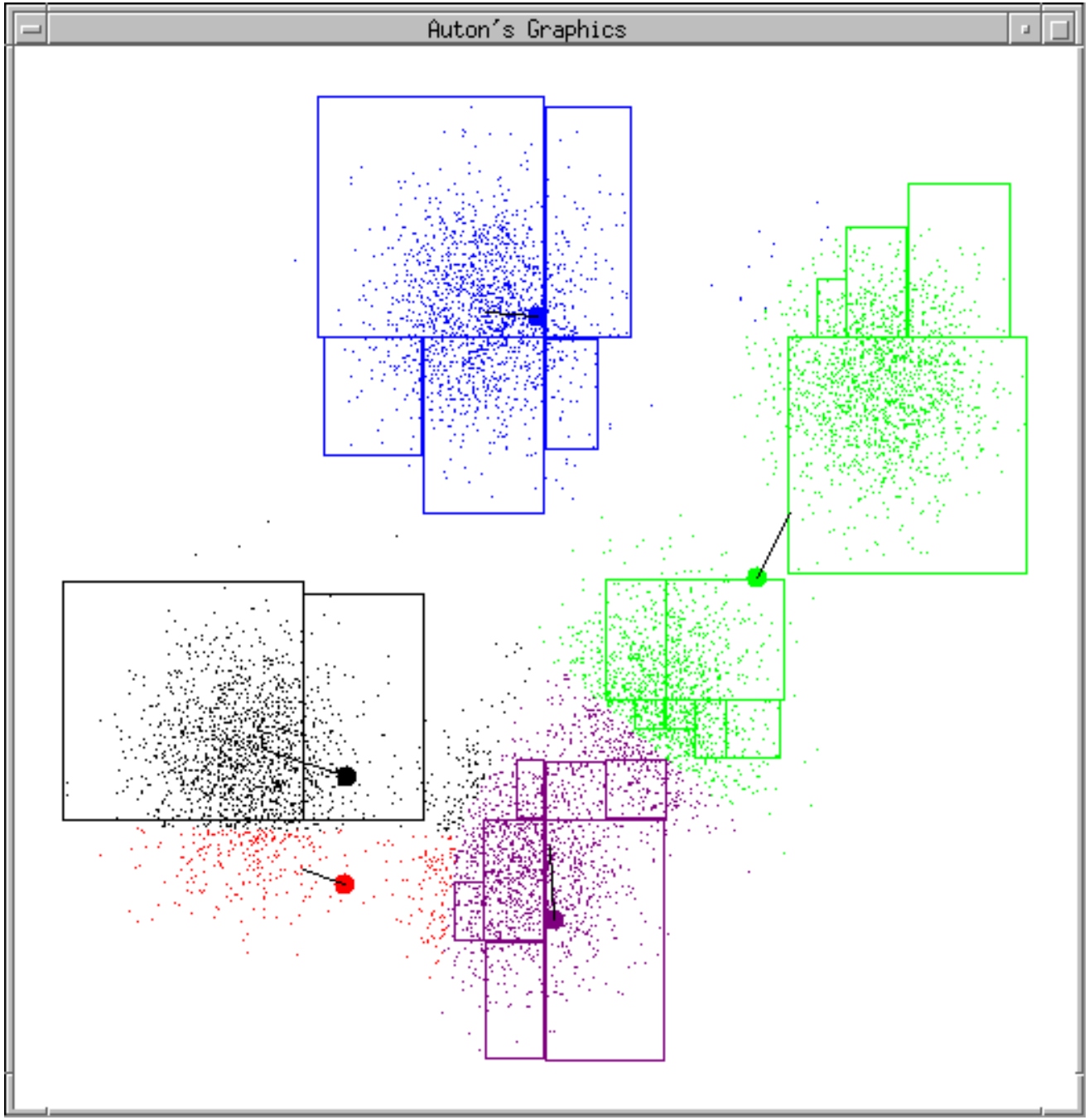
Example generated by Dan Pelleg's super-duper fast K-means system:

Dan Pelleg and Andrew Moore. Accelerating Exact k-means Algorithms with Geometric Reasoning. Proc. Conference on Knowledge Discovery in Databases 1999, (KDD99) (available on www.autonlab.org/pap.html*)*



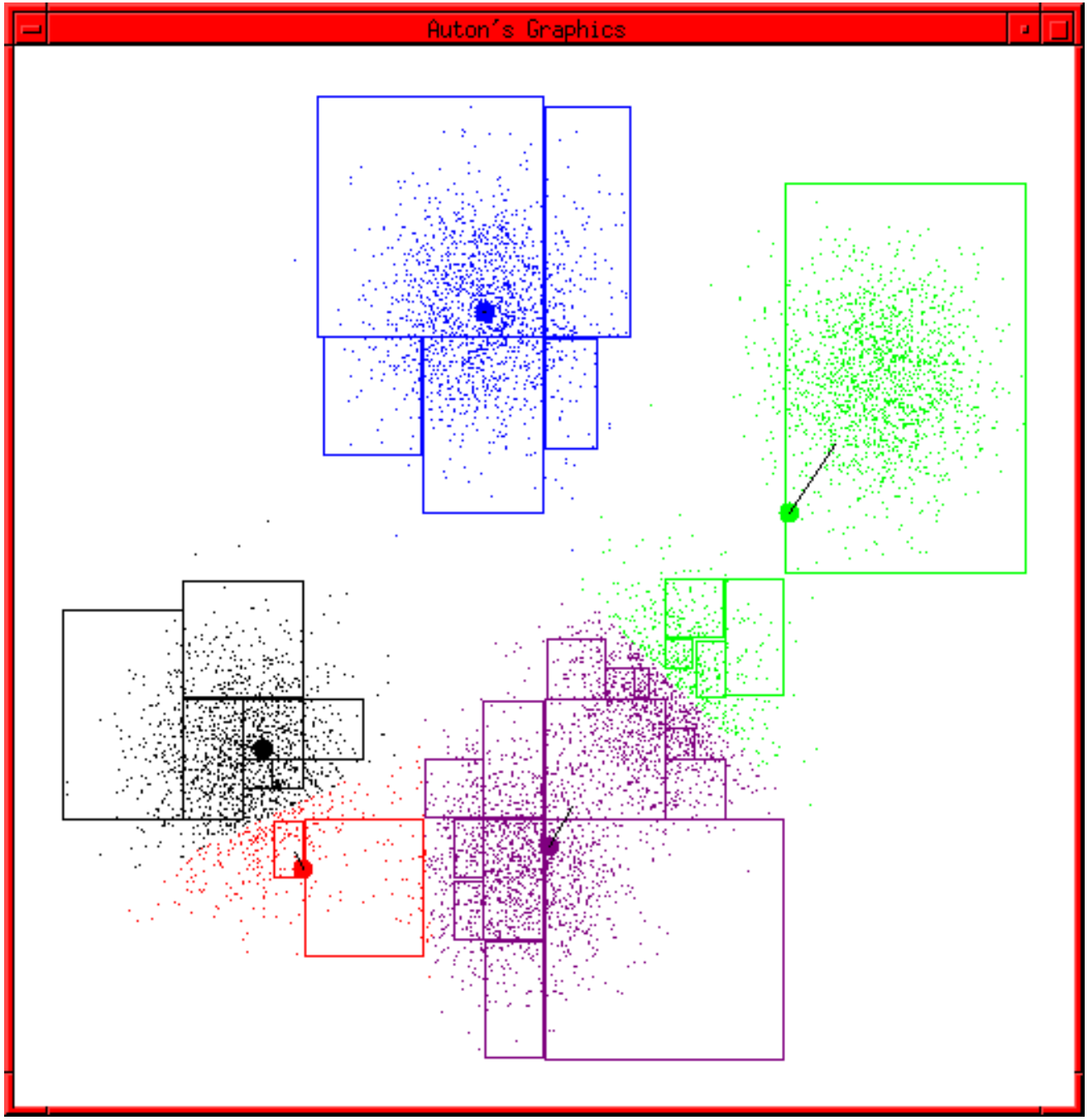
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K-means continues ...



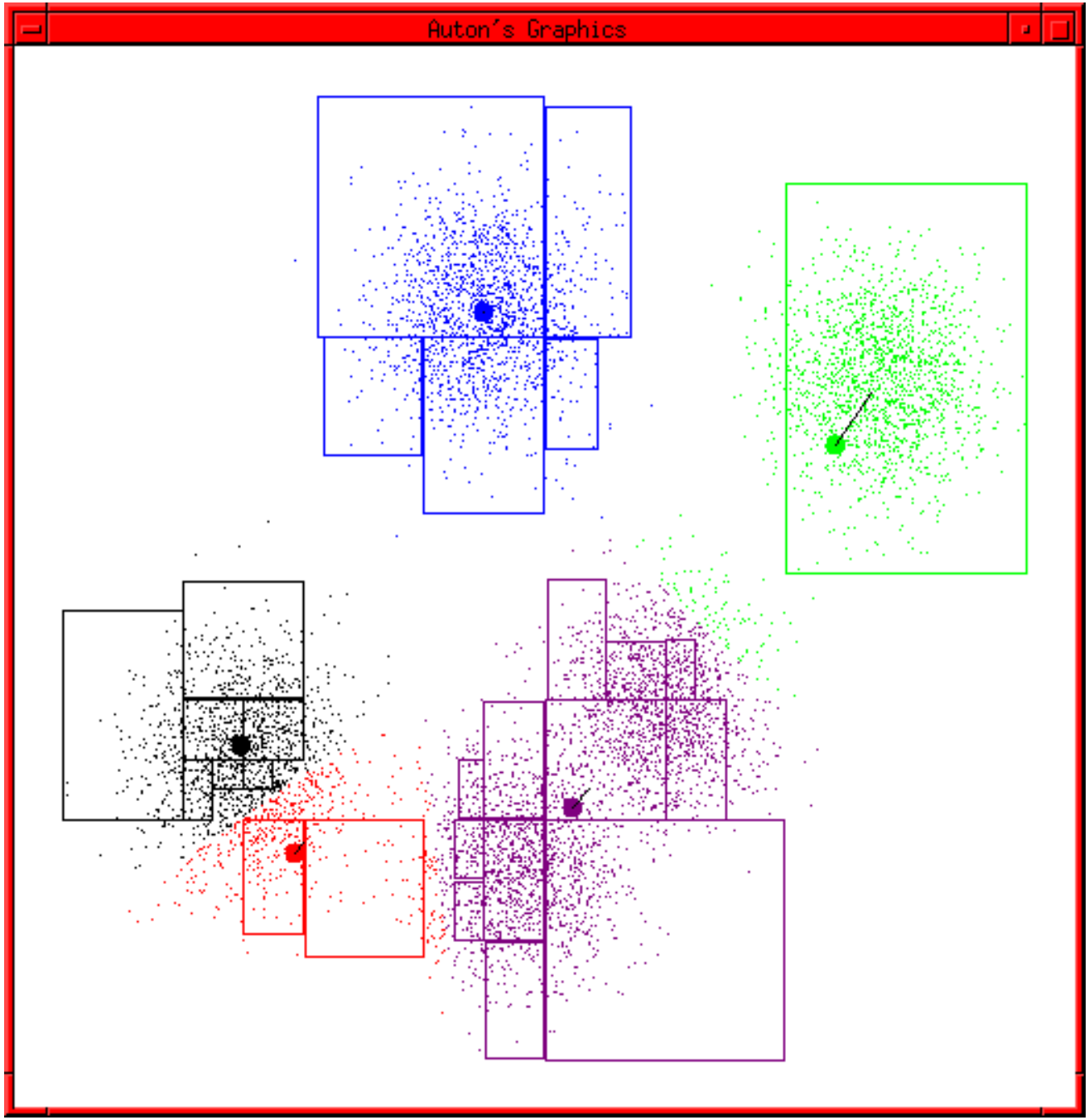
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K-means continues ...



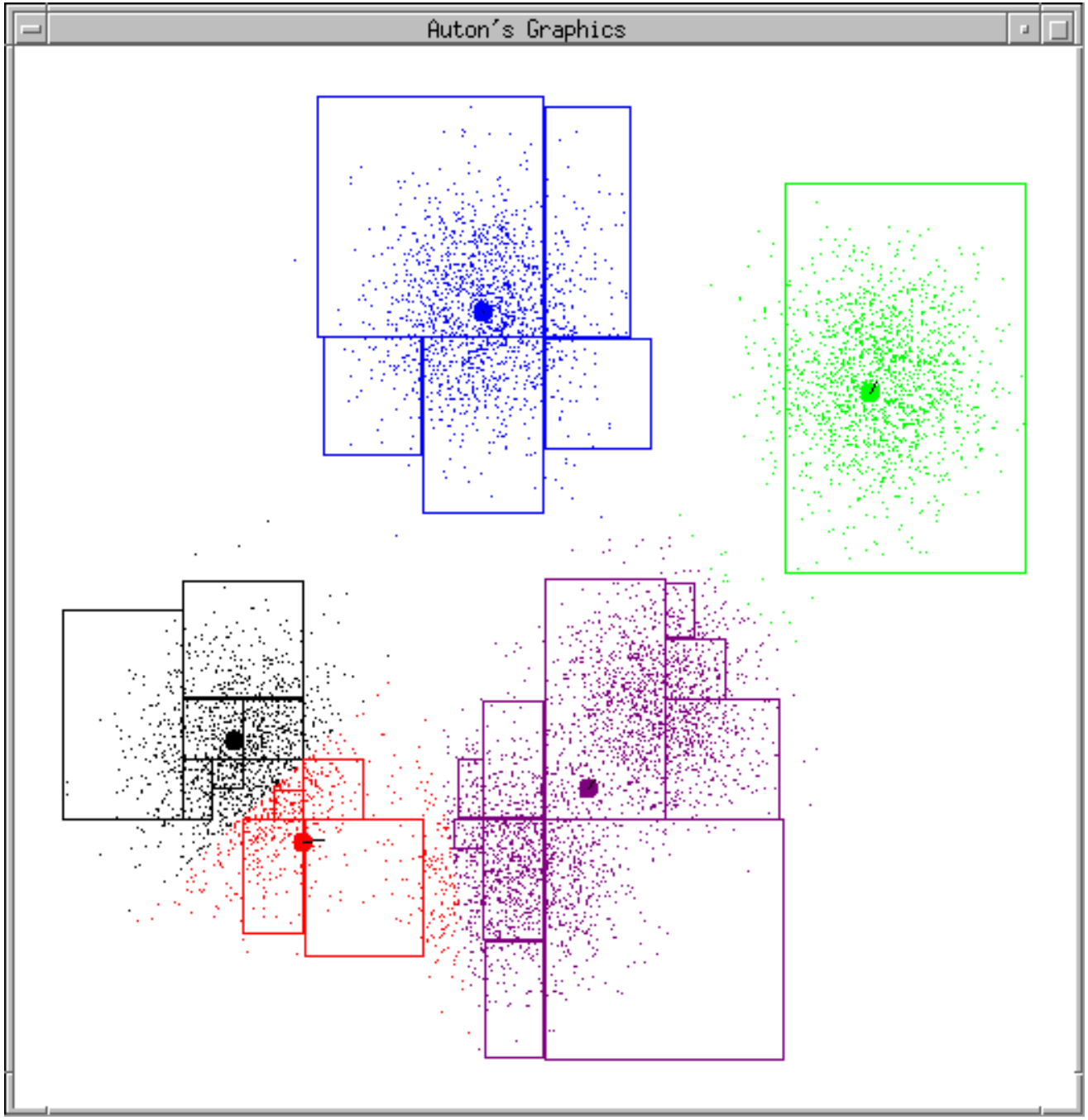
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K-means continues ...



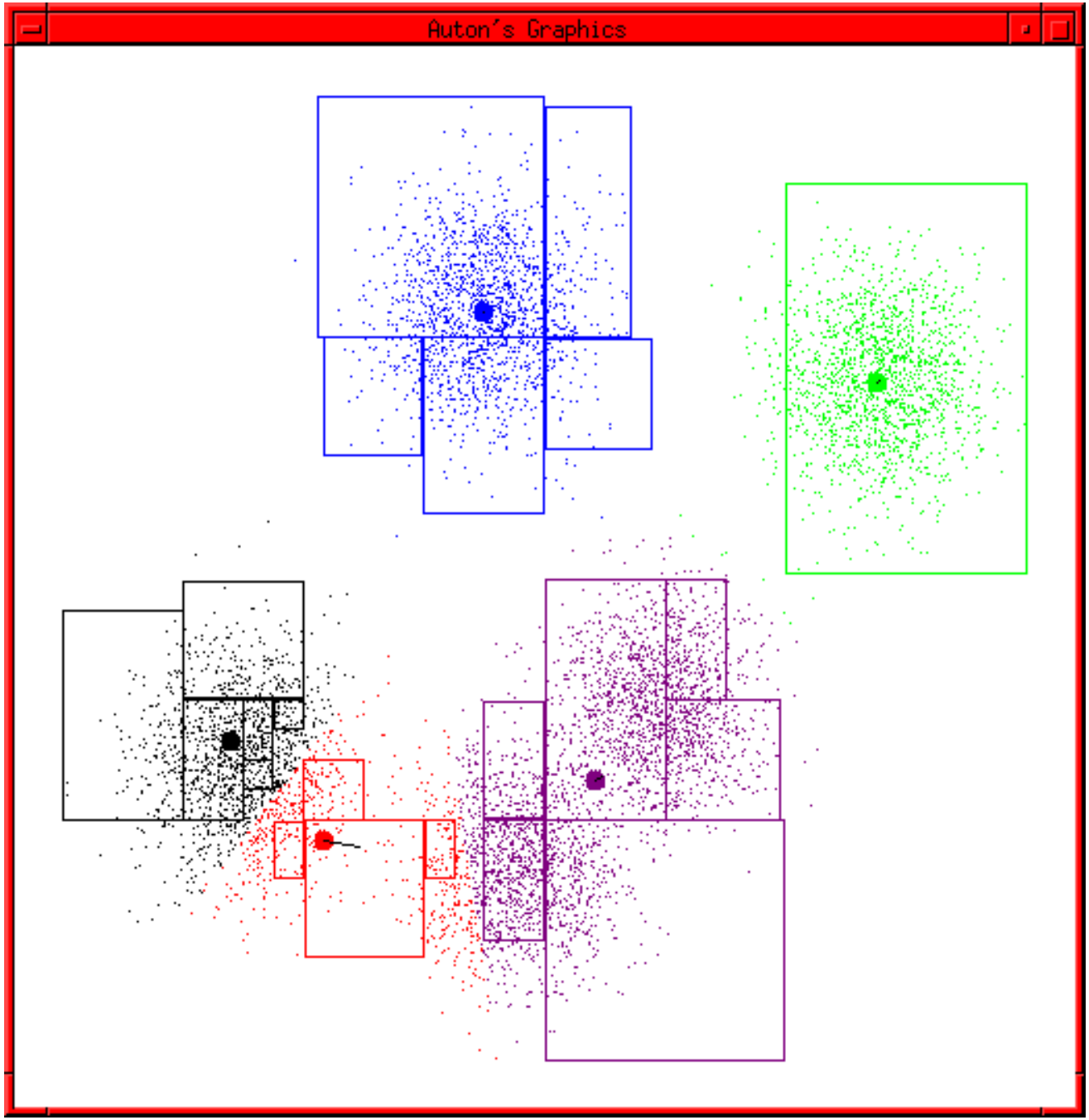
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K-means continues ...



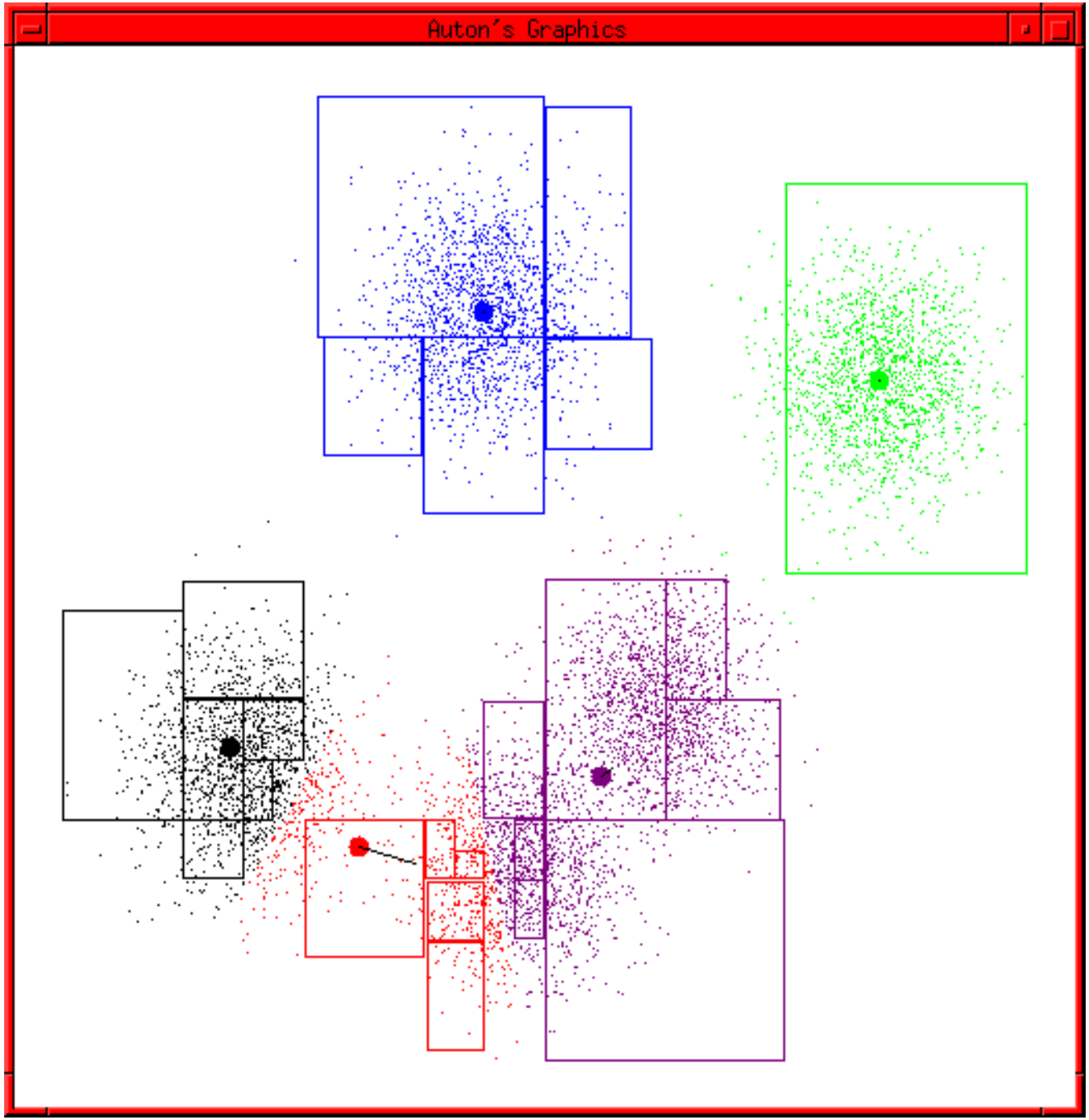
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K-means continues ...



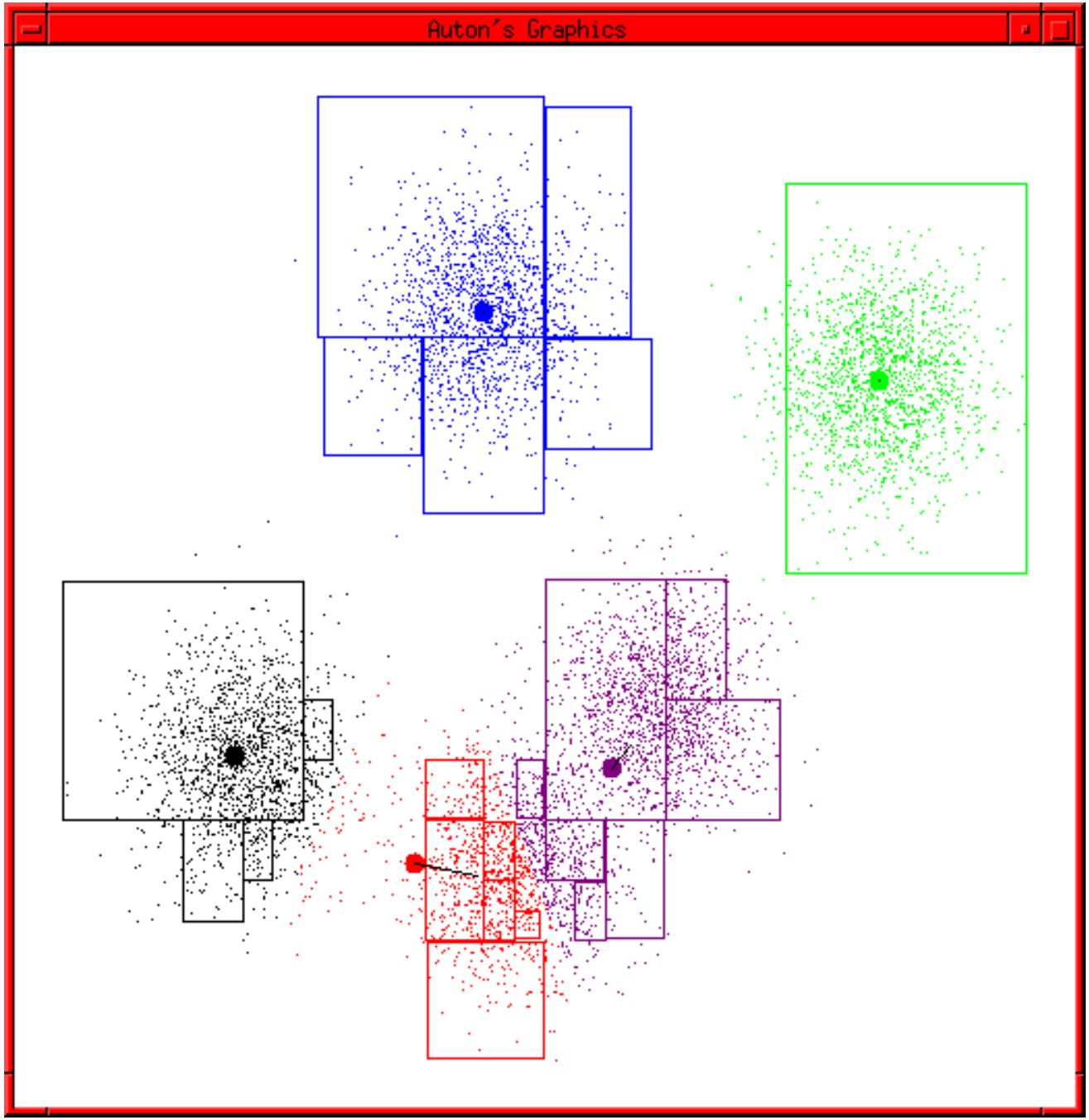
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K-means continues ...



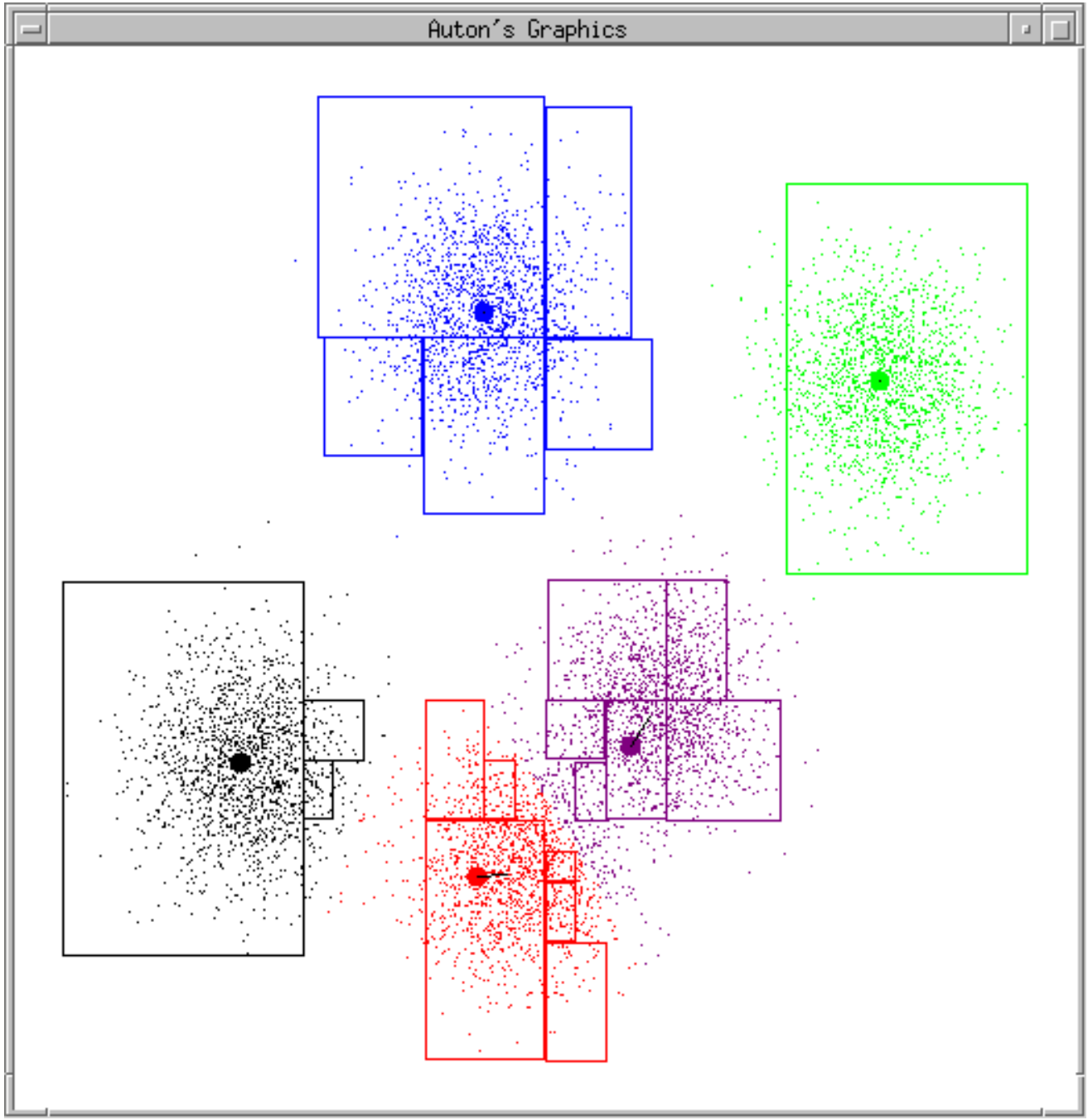
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K-means continues ...



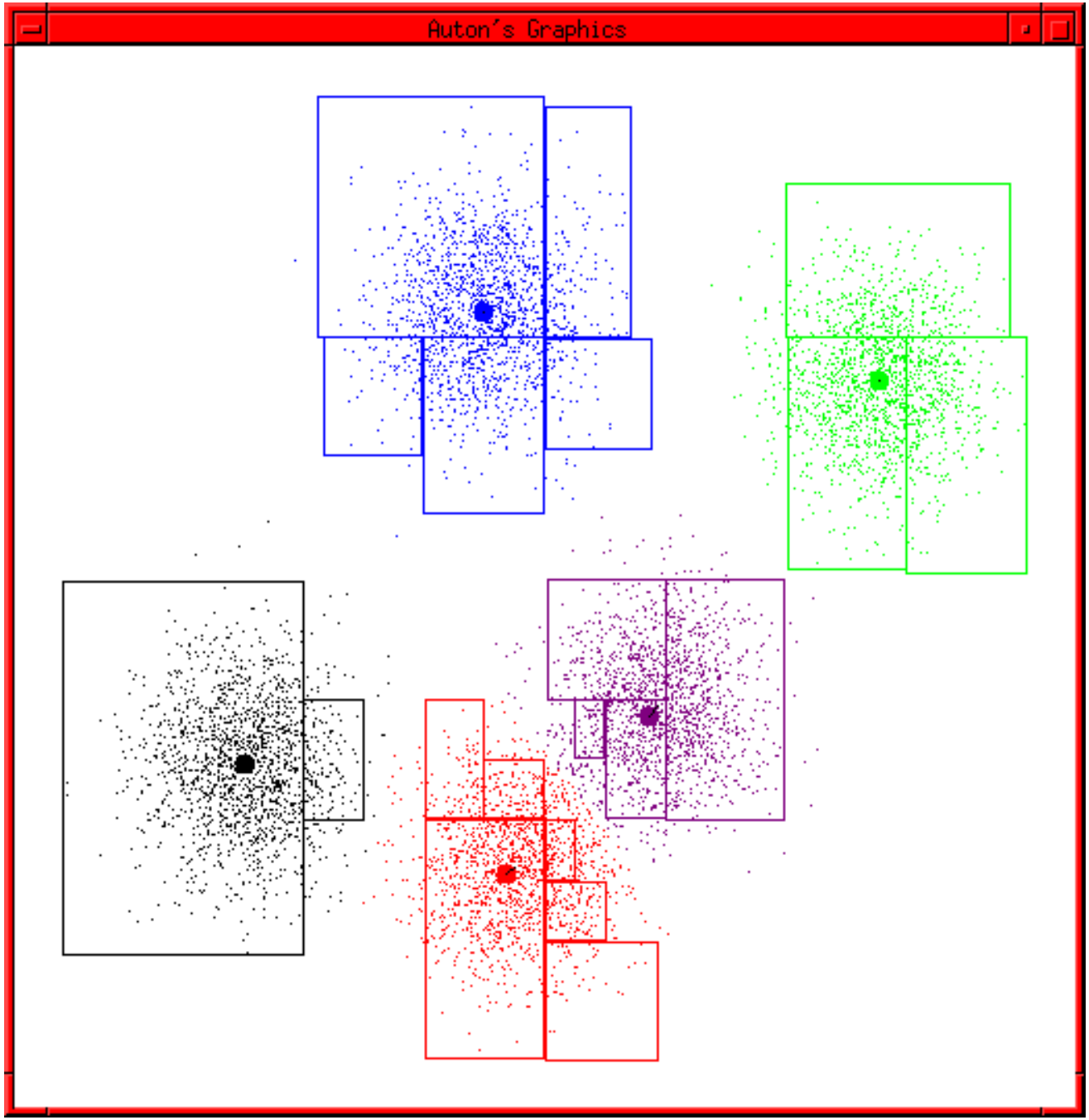
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K-means continues ...



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K-means terminates



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K-means Questions

- What is it trying to optimize?
- Are we sure it will terminate?
- Are we sure it will find an optimal clustering?
- How should we start it?
- How could we automatically choose the number of centers?

....we'll deal with these questions over the next few slides

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Distortion

Given..

•an encoder function: ENCODE : $\Re^m \rightarrow$ [1..k]

•a decoder function: DECODE : [1..k] $\rightarrow \Re^m$

Define...

$$\text{Distortion} = \sum_{i=1}^{R} \left(\mathbf{x}_i - \text{DECODE}[\text{ENCODE}(\mathbf{x}_i)]\right)^2$$

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Distortion

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Define...

$$\text{Distortion} = \sum_{i=1}^{R} \left(\mathbf{x}_i - \text{DECODE}[\text{ENCODE}(\mathbf{x}_i)]\right)^2$$

We may as well write

$$\text{DECODE}[j] = \mathbf{c}_j$$

$$\text{so} \quad \text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

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The Minimal Distortion

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties must centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, ... , $\boldsymbol{c}_k$ have when distortion is minimized?

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The Minimal Distortion (1)

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties must centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, ... , $\boldsymbol{c}_k$ have when distortion is minimized?

(1) $\boldsymbol{x}_i$ must be encoded by its nearest center

....why?

$$\mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)} = \arg\min_{\mathbf{c}_j \in \{\mathbf{c}_1, \mathbf{c}_2, \ldots \mathbf{c}_k\}} (\mathbf{x}_i - \mathbf{c}_j)^2$$

..at the minimal distortion

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The Minimal Distortion (1)

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties must centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, … , $\boldsymbol{c}_k$ have when distortion is minimized?

(1) $\boldsymbol{x}_i$ must be encoded by its nearest center

….why?

Otherwise distortion could be reduced by replacing ENCODE[$\boldsymbol{x}_i$] by the nearest center

$$\mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)} = \arg\min_{\mathbf{c}_j \in \{\mathbf{c}_1, \mathbf{c}_2, \ldots \mathbf{c}_k\}} (\mathbf{x}_i - \mathbf{c}_j)^2$$

..at the minimal distortion

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The Minimal Distortion (2)

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties must centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, ... , $\boldsymbol{c}_k$ have when distortion is minimized?

(2) The partial derivative of Distortion with respect to each center location must be zero.

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(2) The partial derivative of Distortion with respect to each center location must be zero.

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

$$= \sum_{j=1}^{k} \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} (\mathbf{x}_i - \mathbf{c}_j)^2$$

OwnedBy($\boldsymbol{c}_j$) = the set of records owned by Center $\boldsymbol{c}_j$.

$$\frac{\partial \text{Distortion}}{\partial \mathbf{c}_j} = \frac{\partial}{\partial \mathbf{c}_j} \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} (\mathbf{x}_i - \mathbf{c}_j)^2$$

$$= -2 \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} (\mathbf{x}_i - \mathbf{c}_j)$$

$$= 0 \text{ (for a minimum)}$$

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(2) The partial derivative of Distortion with respect to each center location must be zero.

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

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$$\frac{\partial \text{Distortion}}{\partial \mathbf{c}_j} = \frac{\partial}{\partial \mathbf{c}_j} \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} (\mathbf{x}_i - \mathbf{c}_j)^2$$

$$= -2 \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} (\mathbf{x}_i - \mathbf{c}_j)$$

$$= 0 \text{ (for a minimum)}$$

Thus, at a minimum: $\mathbf{c}_j = \frac{1}{|\text{OwnedBy}(\mathbf{c}_j)|} \sum_{i \in \text{OwnedBy}(\mathbf{c}_j)} \mathbf{x}_i$

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At the minimum distortion

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties must centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, ... , $\boldsymbol{c}_k$ have when distortion is minimized?

(1) $\boldsymbol{x}_i$ must be encoded by its nearest center

(2) Each Center must be at the centroid of points it owns.

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Improving a suboptimal configuration...

$$\text{Distortion} = \sum_{i=1}^{R} (\mathbf{x}_i - \mathbf{c}_{\text{ENCODE}(\mathbf{x}_i)})^2$$

What properties can be changed for centers $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, ... , $\boldsymbol{c}_k$ have when distortion is not minimized?

(1) Change encoding so that $\boldsymbol{x}_i$ is encoded by its nearest center

(2) Set each Center to the centroid of points it owns.

There's no point applying either operation twice in succession.

But it can be profitable to alternate.

...And that's K-means!

Easy to prove this procedure will terminate in a state at which neither (1) or (2) change the configuration. Why?

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Improving a suboptimal configuration...

What p… …c_k
have w…

(1) Cha… …center

(2) Set …

There's …ssion.

But it ca… be profitable to alternate.

...And that's K-means!

There are only a finite number of ways of partitioning R records into k groups.

So there are only a finite number of possible configurations in which all Centers are the centroids of the points they own.

If the configuration changes on an iteration, it must have improved the distortion.

So each time the configuration changes it must go to a configuration it's never been to before.

So if it tried to go on forever, it would eventually run out of configurations.

Easy to prove this procedure will terminate in a state at which neither (1) or (2) change the configuration. Why?

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K-Means

Algorithm

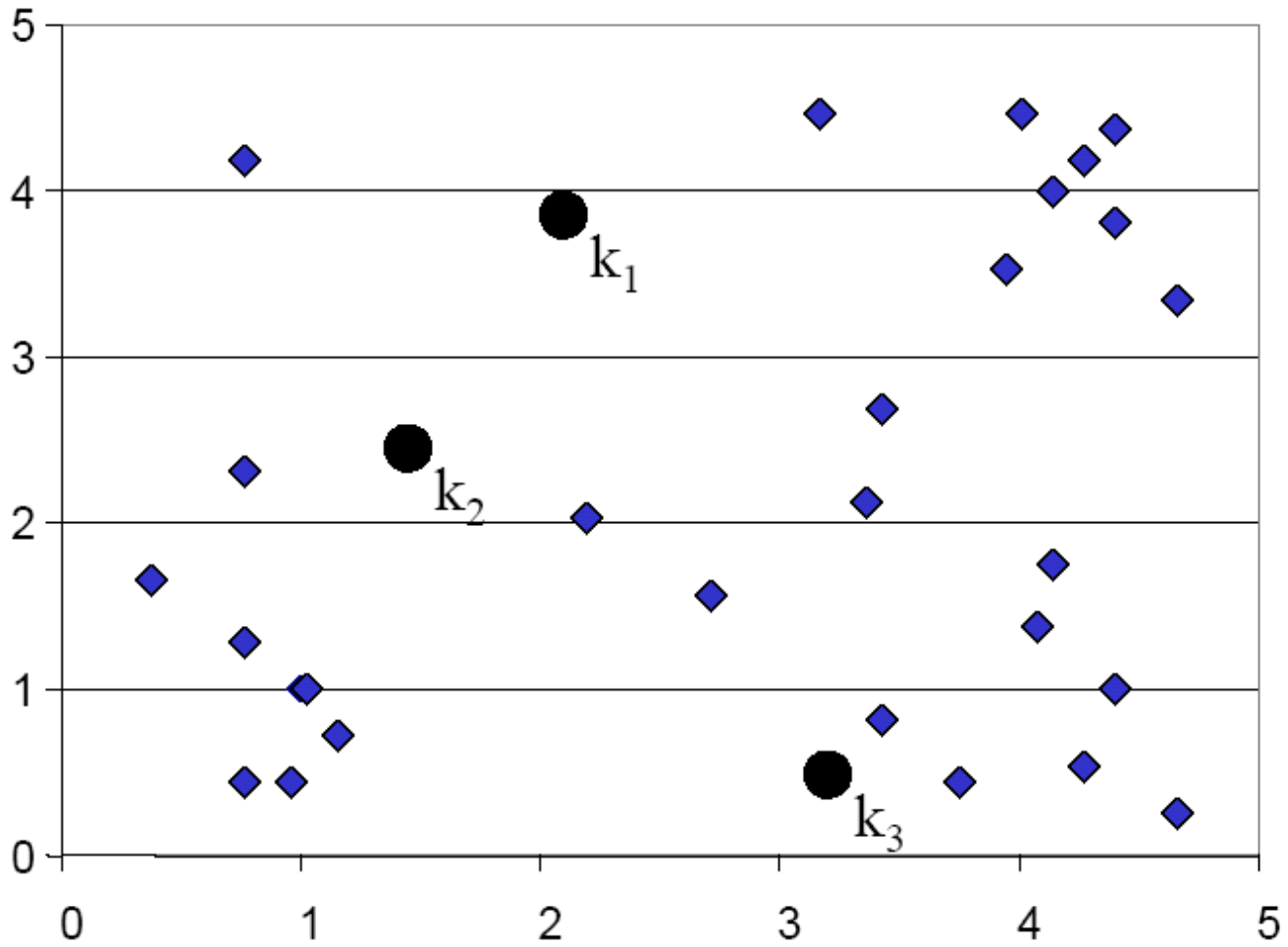
1. Decide on a value for *K*.
2. Initialize the *K* cluster centers randomly if necessary.
3. Decide the **class memberships** of the *N* objects by assigning them to the nearest cluster centroids (aka the center of gravity or mean)

4. Re-estimate the ***K* cluster centers**, by assuming the memberships found above are correct.

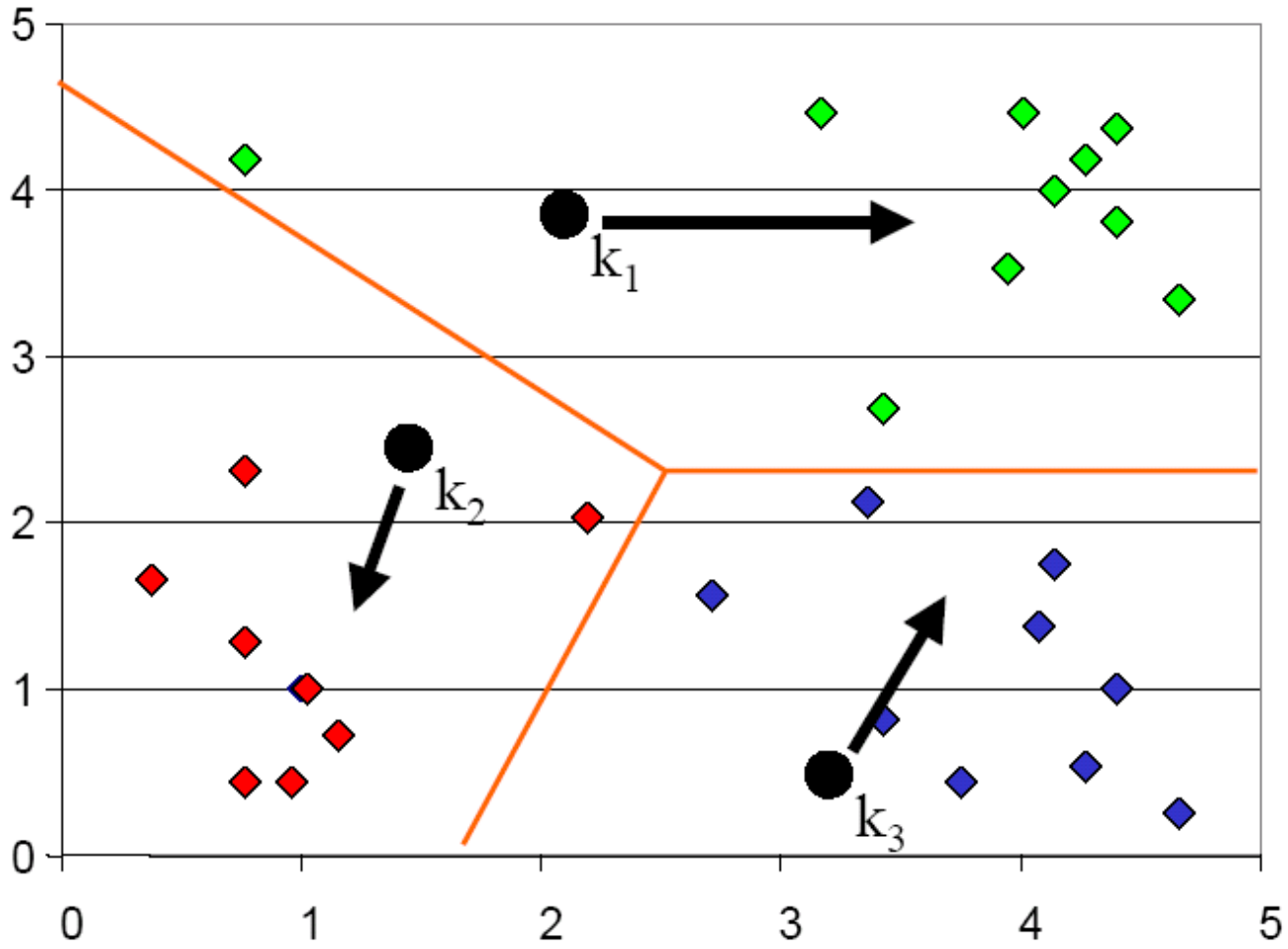
$$\vec{\mu}_k = \frac{1}{\mathcal{C}_k} \sum_{i \in \mathcal{C}_k} \vec{x}_i$$

5. If none of the *N* objects changed membership in the last iteration, exit. Otherwise go to 3.

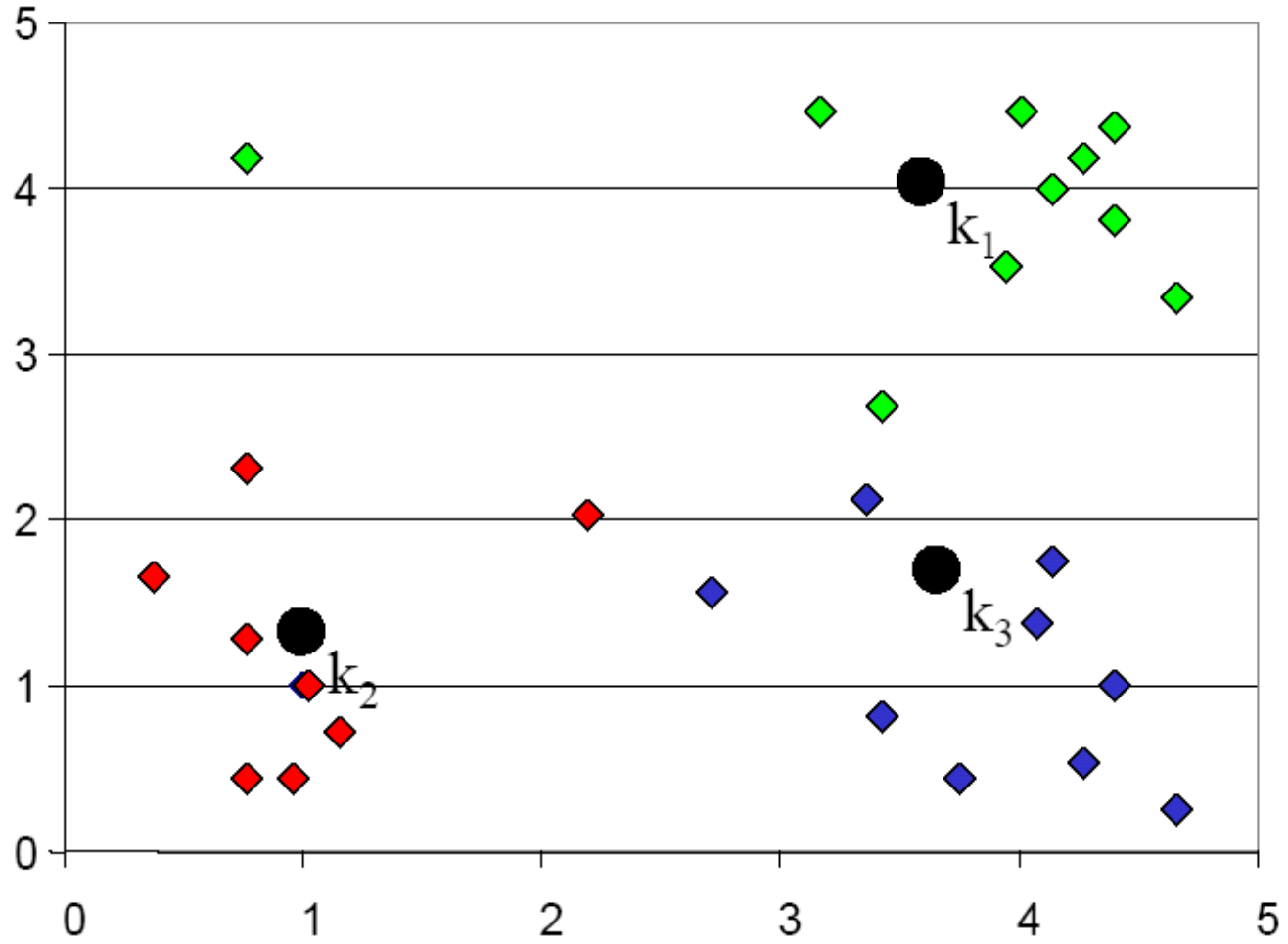
K-means Clustering: Step 1



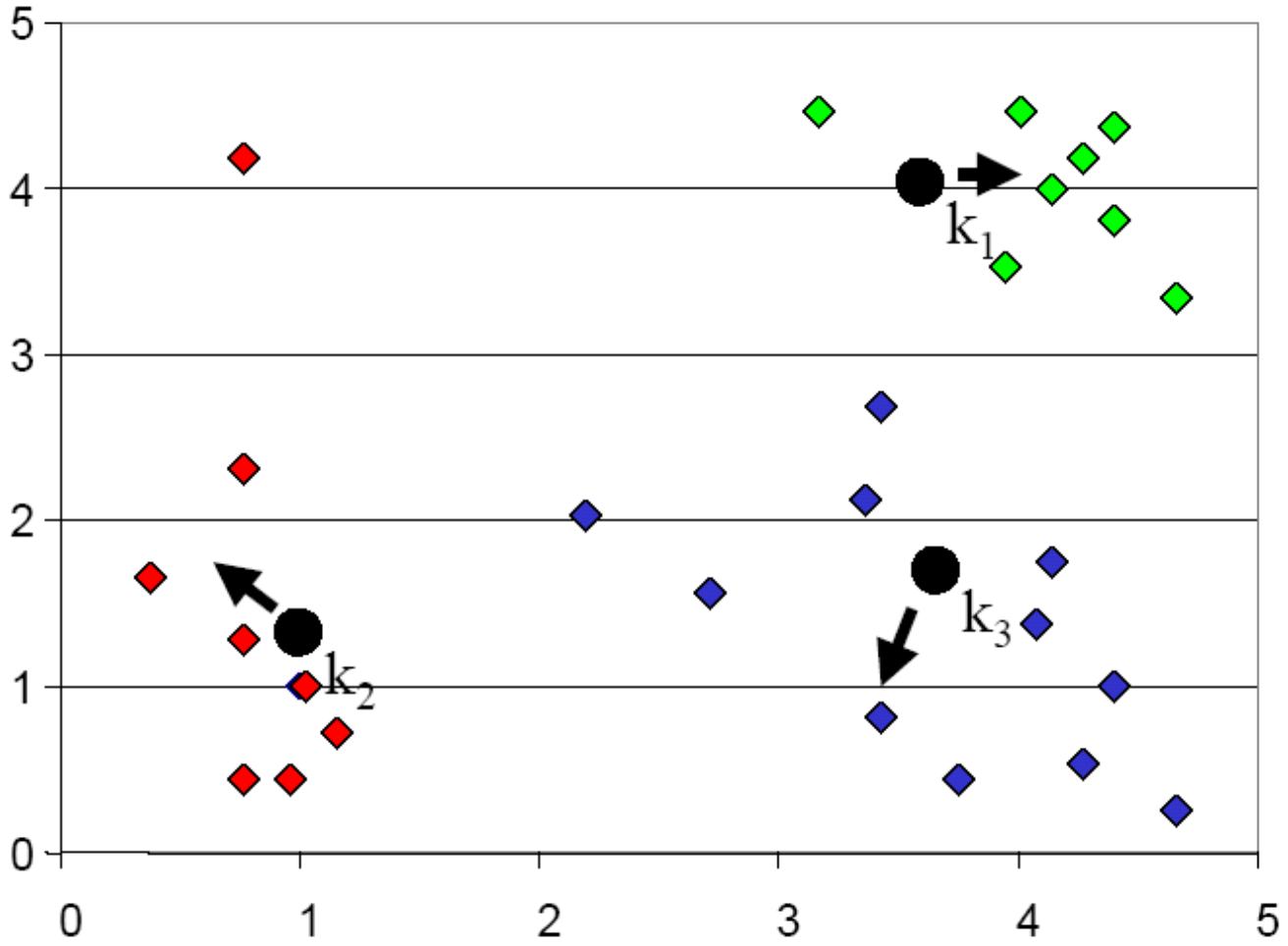
K-means Clustering: Step 2



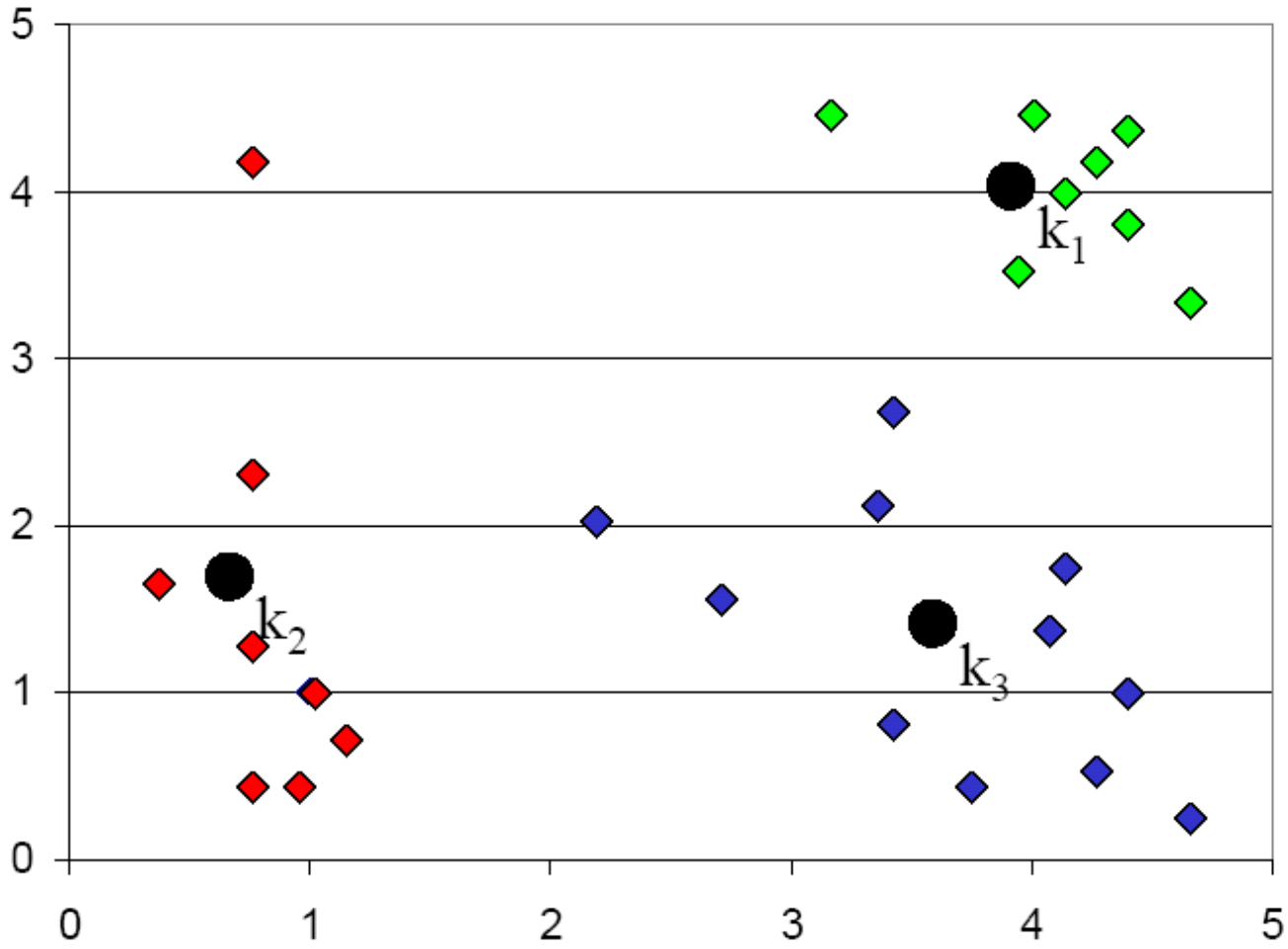
K-means Clustering: Step 3



K-means Clustering: Step 4



K-means Clustering: Step 5



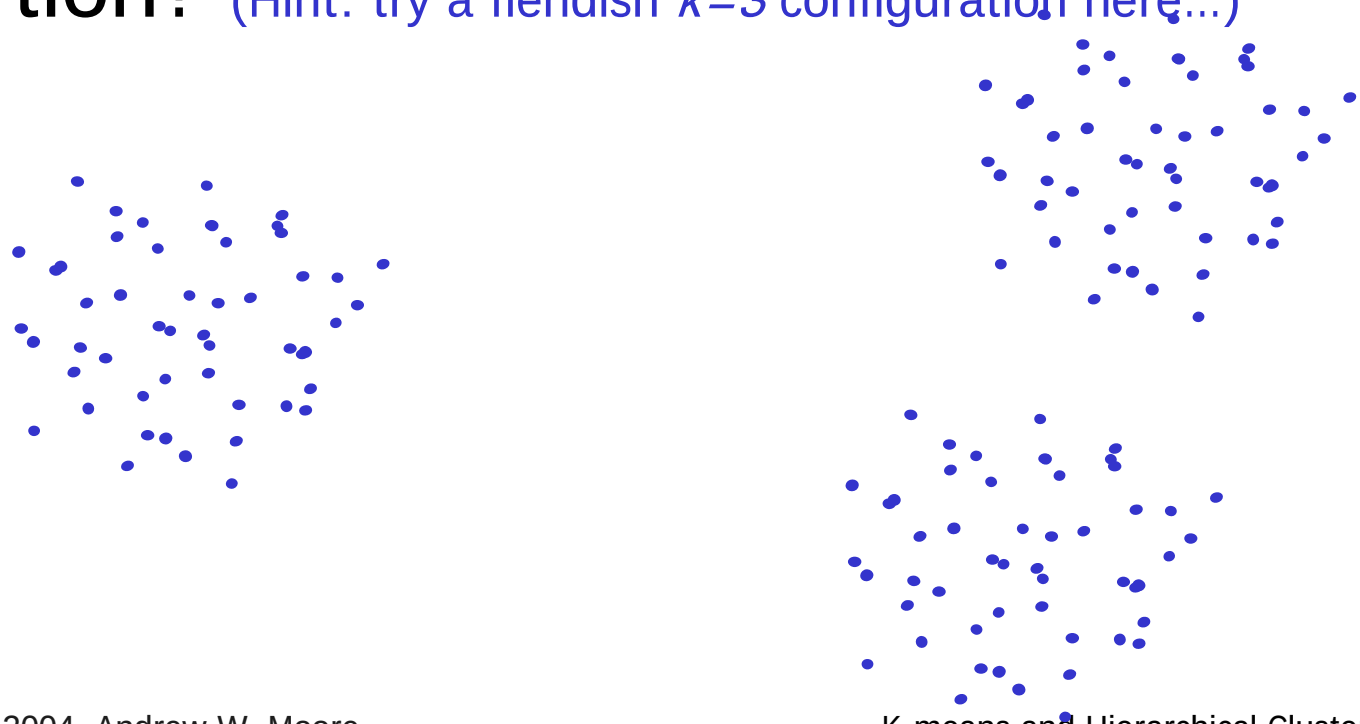
Will we find the optimal configuration?

- Not necessarily.
- Can you invent a configuration that has converged, but does not have the minimum distortion?

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Will we find the optimal configuration?

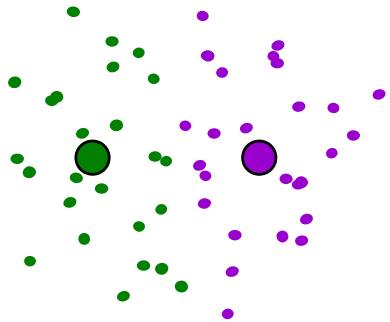
- Not necessarily.
- Can you invent a configuration that has converged, but does not have the minimum distortion? (Hint: try a fiendish *k=3* configuration here...)

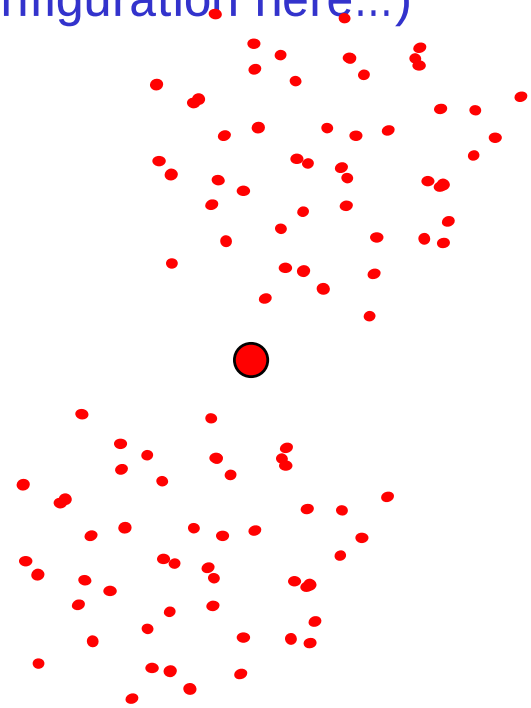


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Will we find the optimal configuration?

- Not necessarily.
- Can you invent a configuration that has converged, but does not have the minimum distortion? (Hint: try a fiendish *k=3* configuration here...)

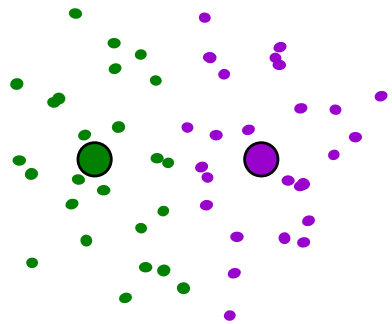


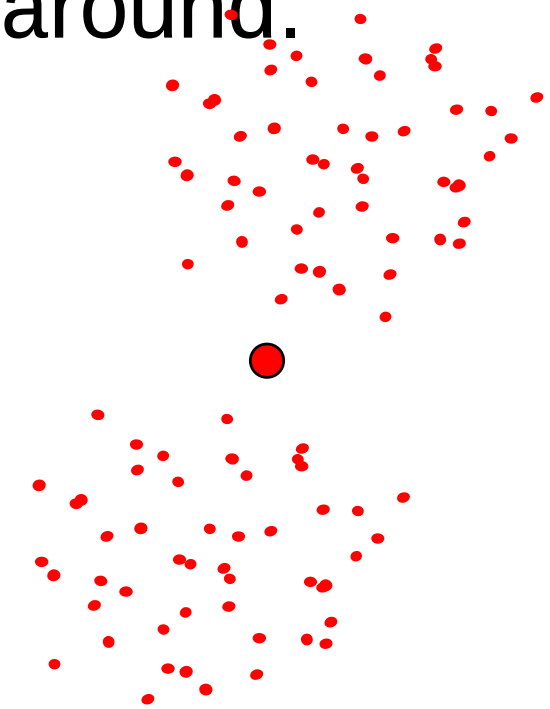


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Trying to find good optima

- Idea 1: Be careful about where you start
- Idea 2: Do many runs of k-means, each from a different random start configuration
- Many other ideas floating around.





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Trying to find good optima

- Idea 1: Be careful about where you start
- Idea 2: D…ny runs of k-means, each fro…
- Mar…

Neat trick:

Place first center on top of randomly chosen datapoint.

Place second center on datapoint that's as far away as possible from first center

:

Place j'th center on datapoint that's as far away as possible from the closest of Centers 1 through j-1

:



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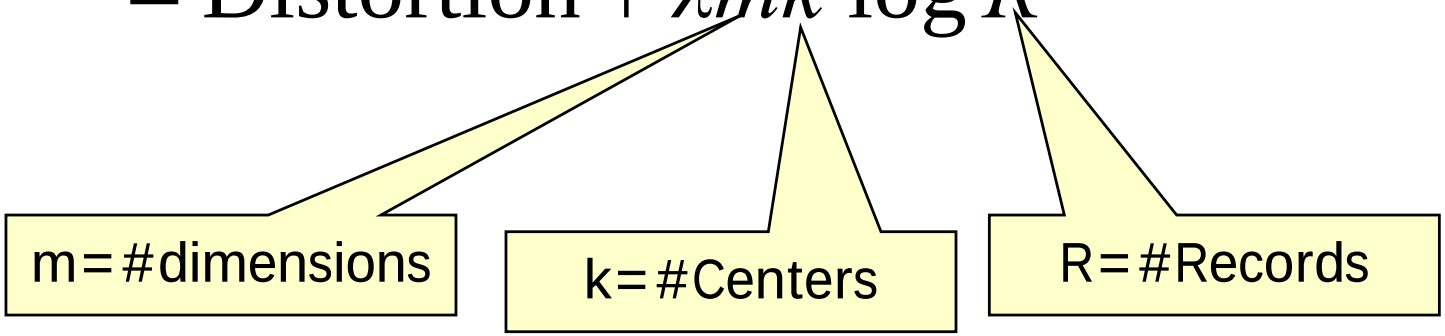
What happens if we choose K according to distortion?

Choosing the number of Centers

- A difficult problem
- Most common approach is to try to find the solution that minimizes the Schwarz Criterion (also related to the BIC)

$$\text{Distortion} + \lambda\,(\#\,\text{parameters})\log R$$

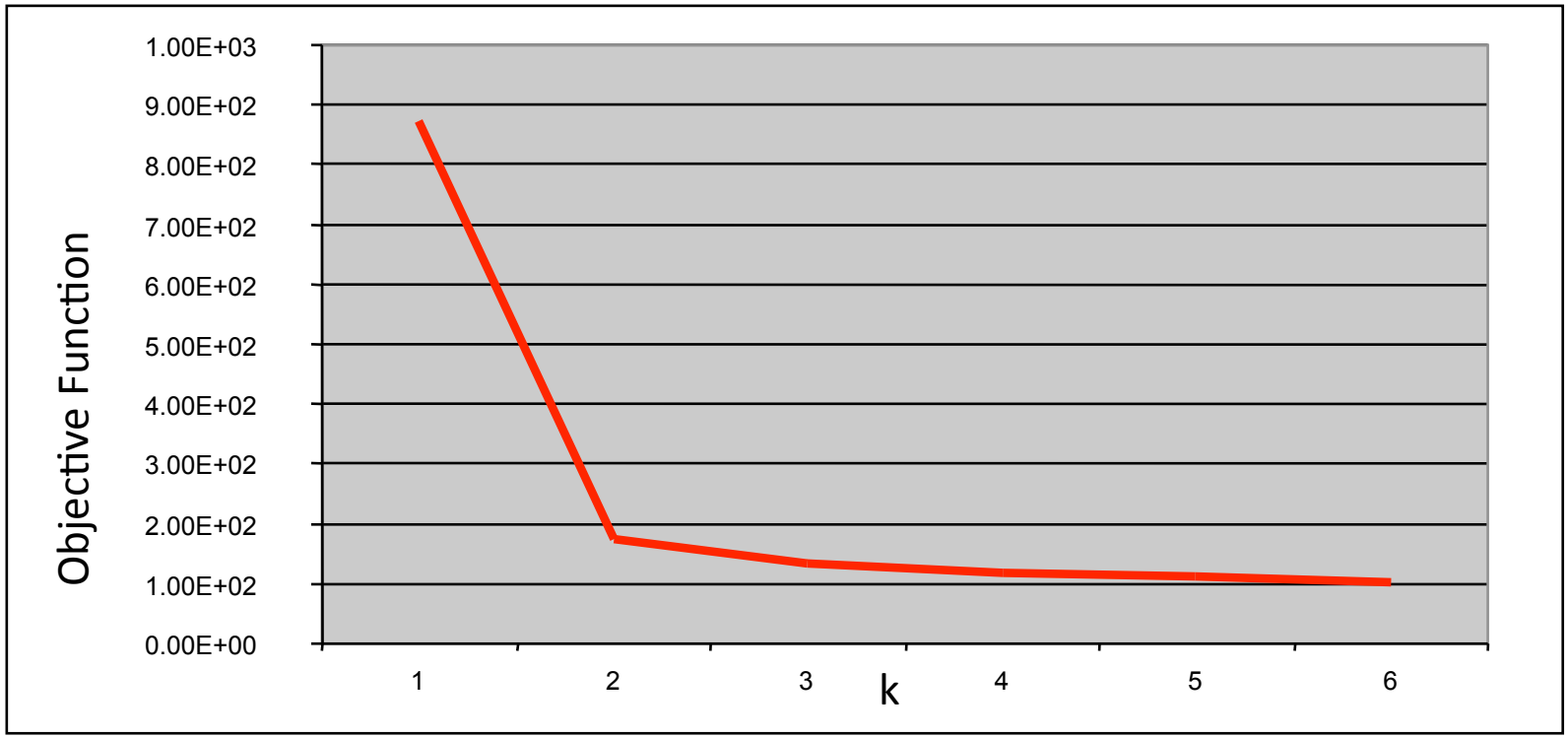
$$= \text{Distortion} + \lambda m k \log R$$



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We can plot the objective function values for k equals 1 to 6…

The abrupt change at k = 2, is highly suggestive of two clusters in the data. This technique for determining the number of clusters is known as “knee finding” or “elbow finding”.



Note that the results are not always as clear cut as in this toy example

Common uses of K-means

- Often used as an exploratory data analysis tool
- In one-dimension, a good way to quantize real-valued variables into k non-uniform buckets
- Used on acoustic data in speech understanding to convert waveforms into one of k categories (known as Vector Quantization)
- Also used for choosing color palettes on old fashioned graphical display devices!

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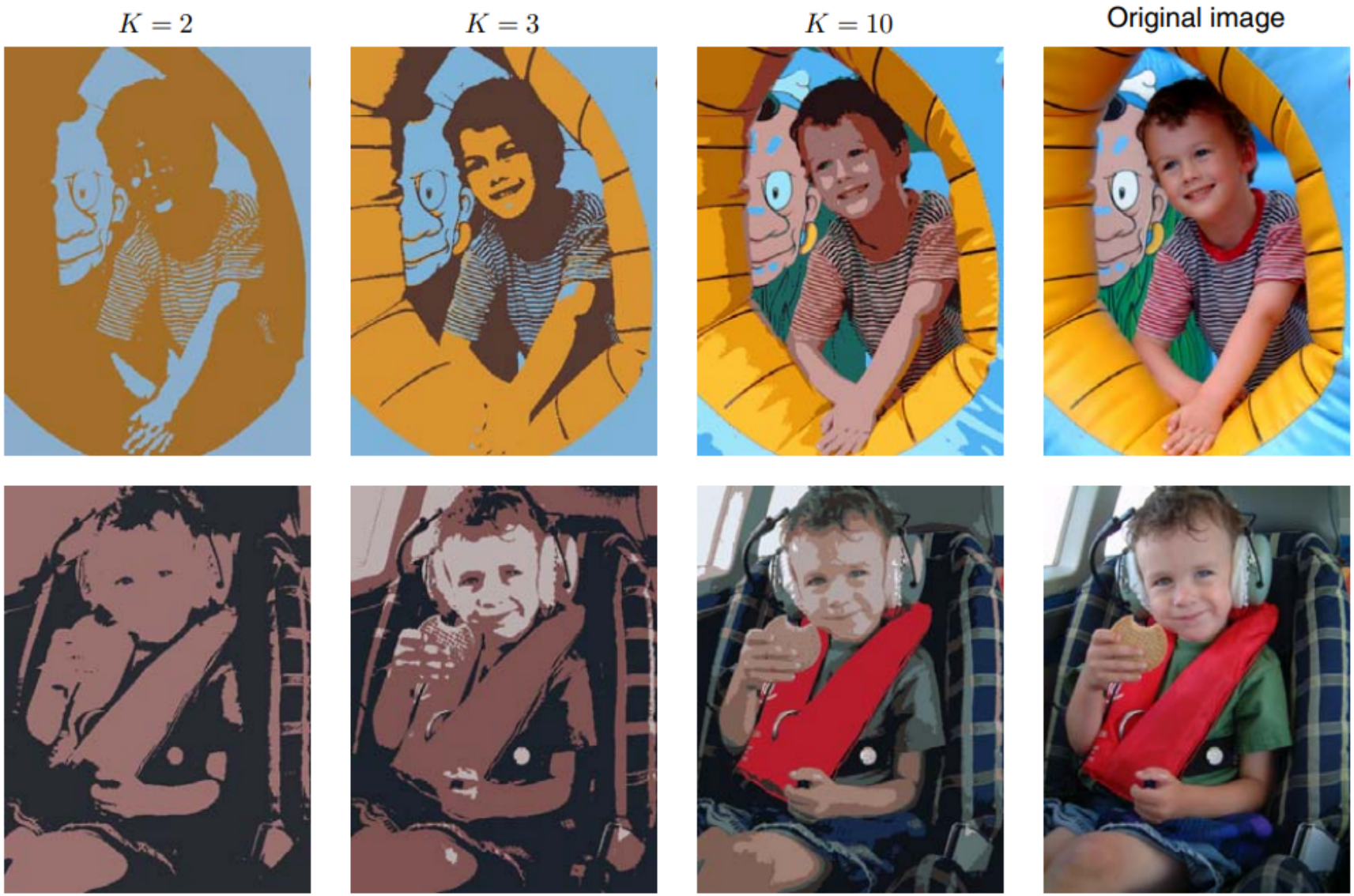


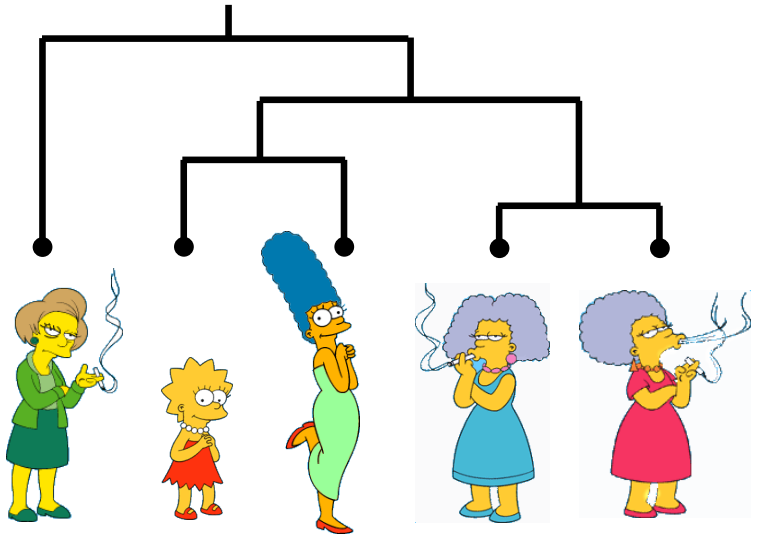
Figure 9.3 Two examples of the application of the K-means clustering algorithm to image segmentation showing the initial images together with their K-means segmentations obtained using various values of K. This also illustrates of the use of vector quantization for data compression, in which smaller values of K give higher compression at the expense of poorer image quality.

Two Types of Clustering

- **Hierarchical algorithms:** Create a hierarchical decomposition of the set of objects using some criterion
- **Partitional algorithms:** Construct various partitions and then evaluate them by some criterion

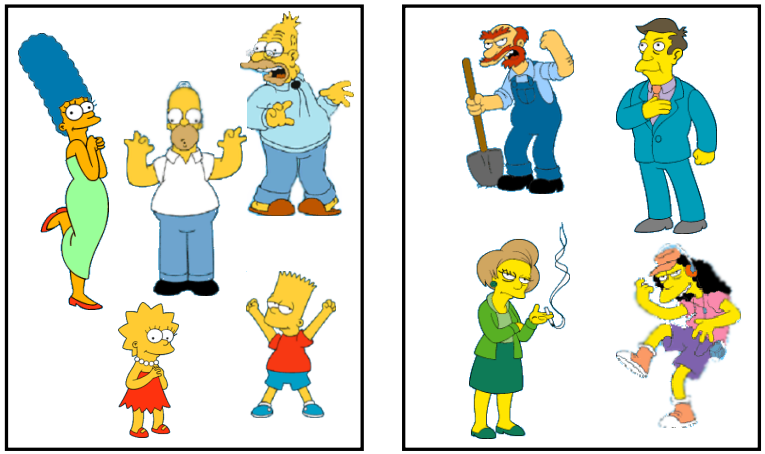
Bottom up or top down

Hierarchical



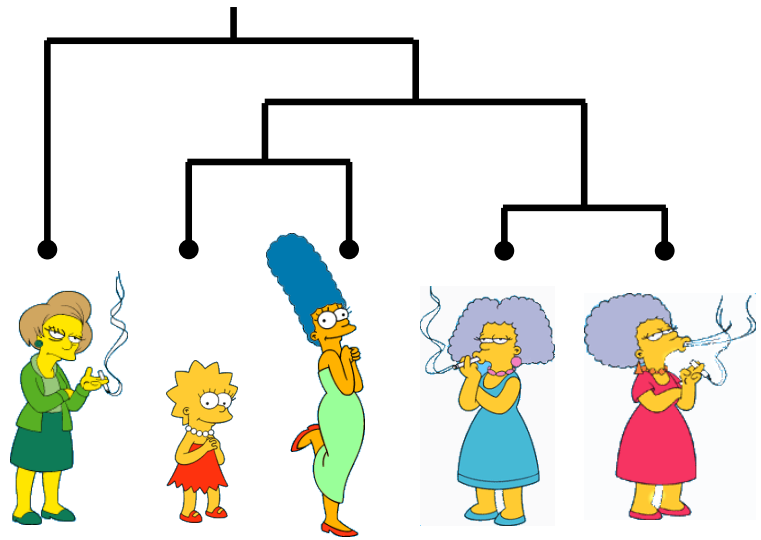
Top down

Partitional



(How-to) Hierarchical Clustering

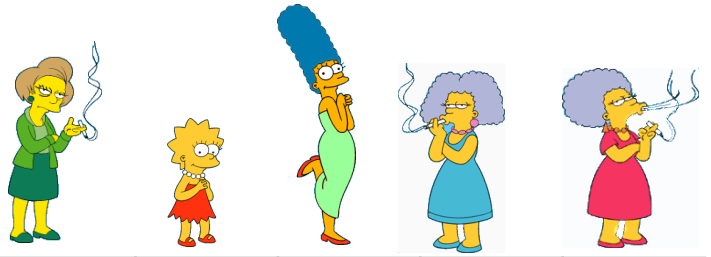
Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.



We begin with a distance matrix which contains the distances between every pair of objects in our database.

D() = 8

D() = 1







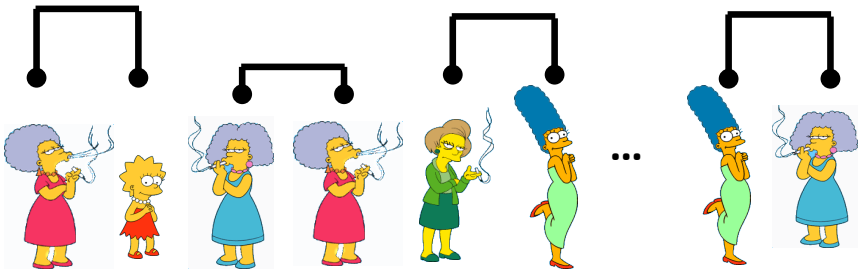




0	8	8	7	7
	0	2	4	4
		0	3	3
			0	1
				0

Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

Consider all possible merges…

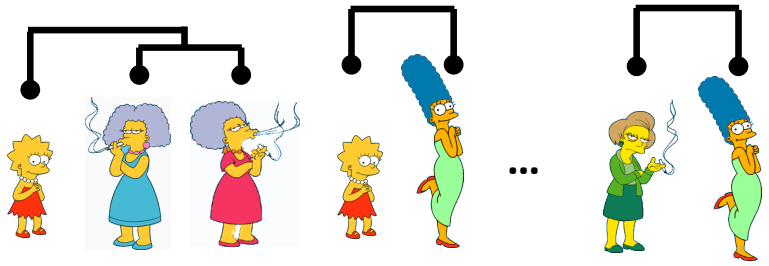


Choose the best

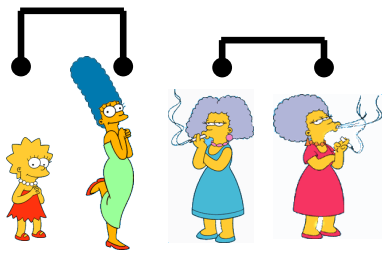


Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

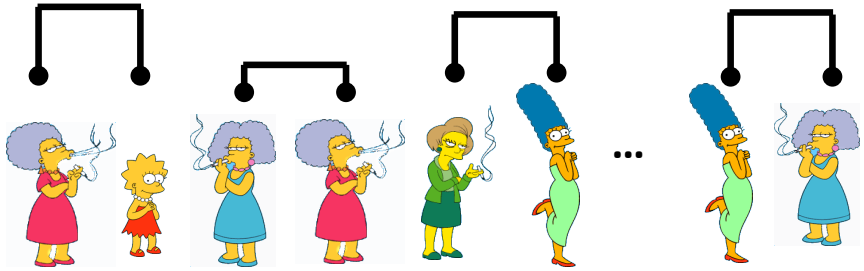
Consider all possible merges…



Choose the best



Consider all possible merges…



Choose the best



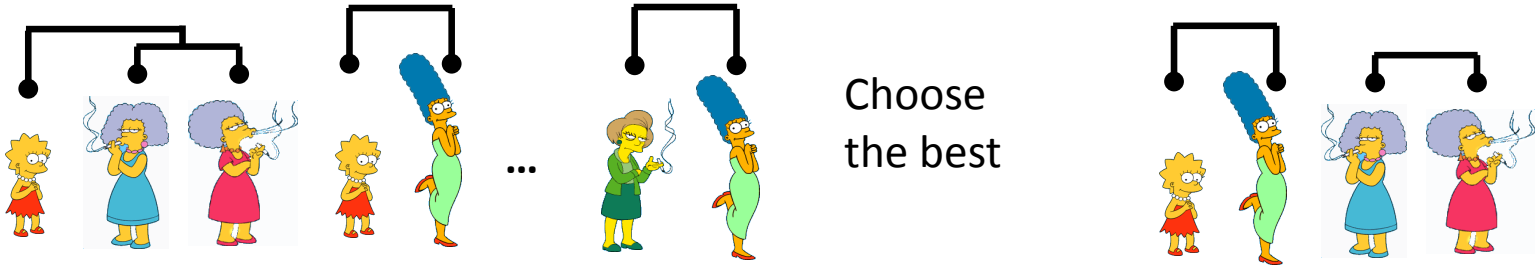
Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

Consider all possible merges…



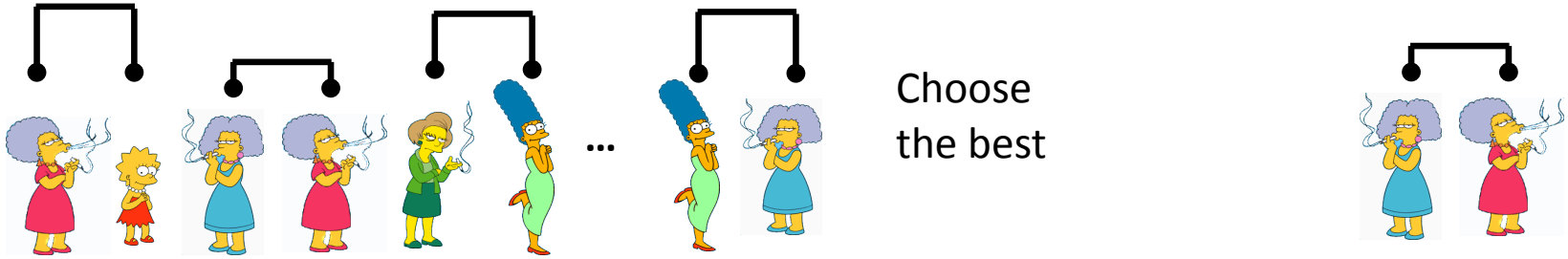
Choose the best

Consider all possible merges…



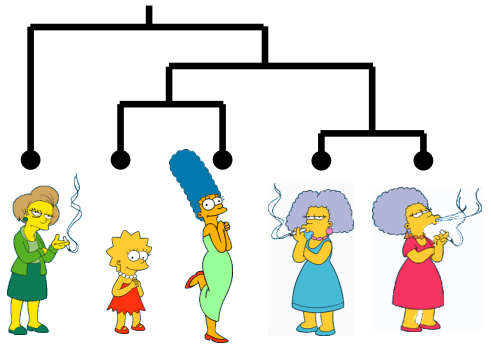
Choose the best

Consider all possible merges…

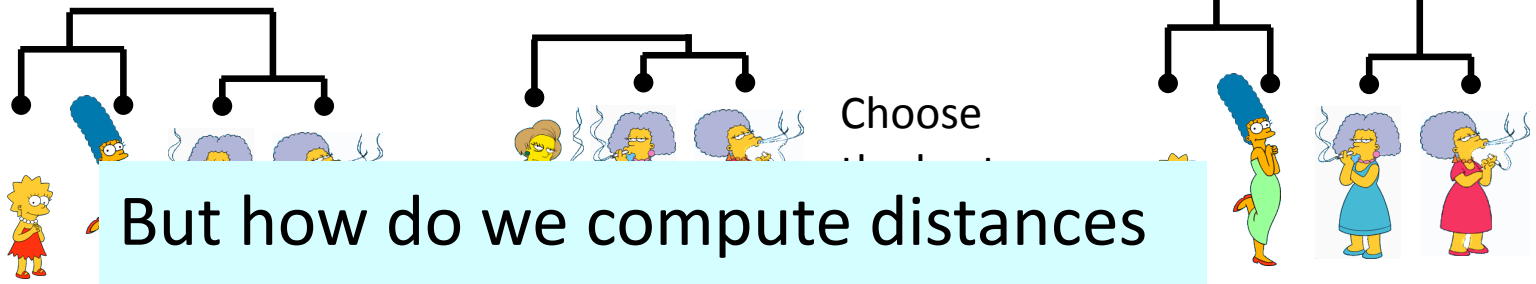


Choose the best

Bottom-Up (agglomerative): Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.



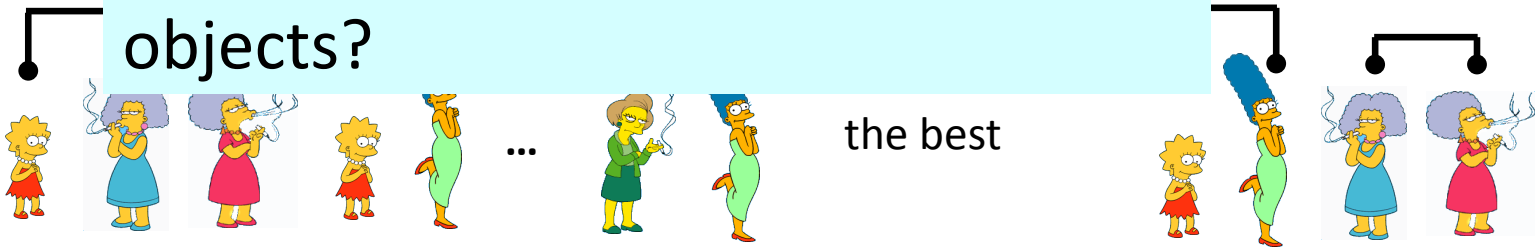
Consider all possible merges…



Choose

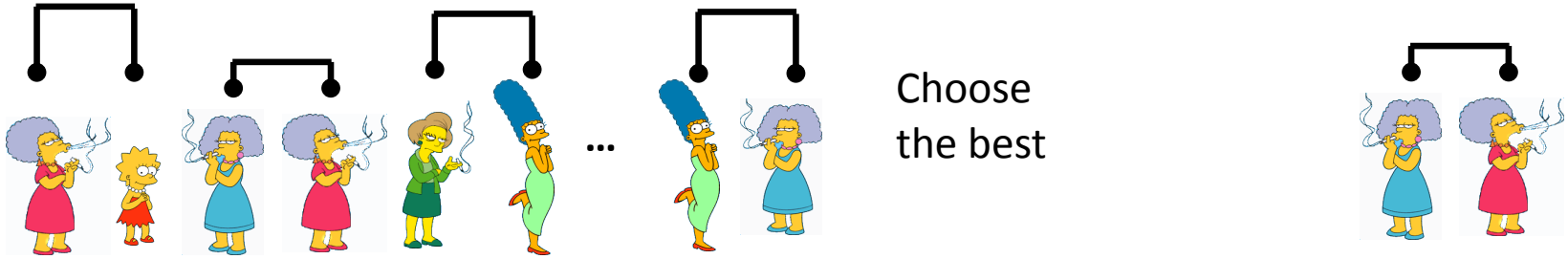
But how do we compute distances between clusters rather than objects?

Consider all possible merges…



the best

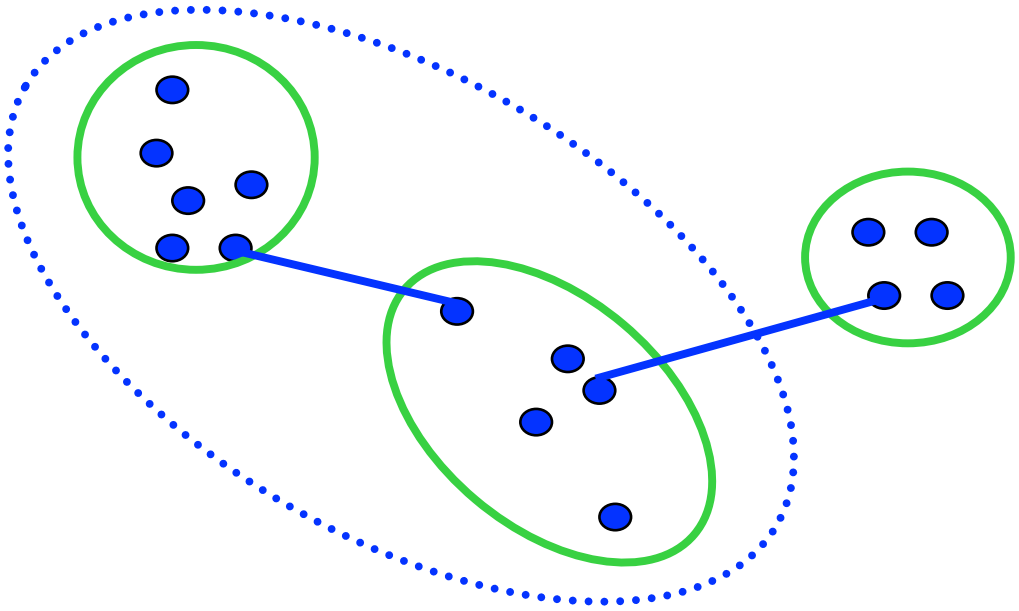
Consider all possible merges…



Choose the best

Computing distance between clusters: Single Link

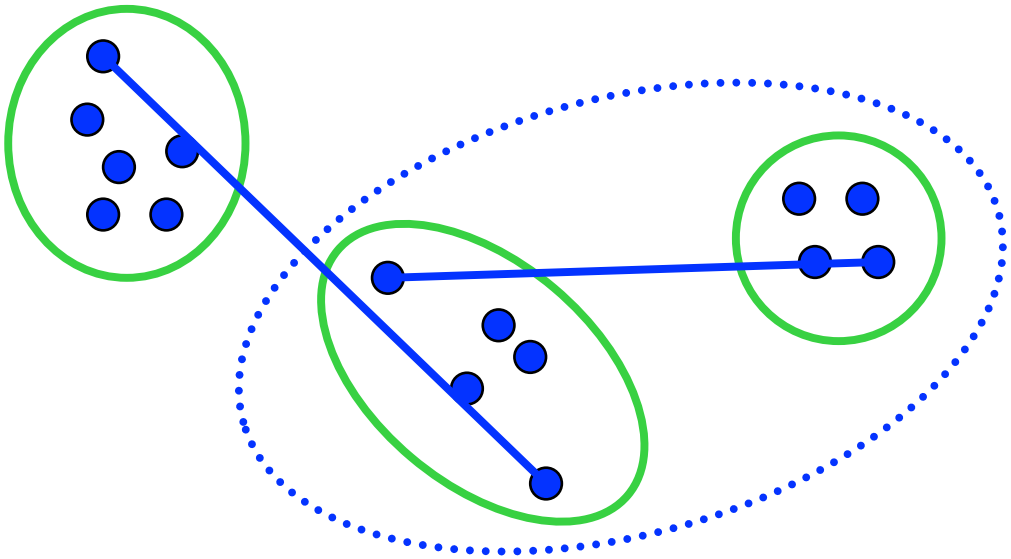
- cluster distance = distance of two closest members in each class



- Potentially long and skinny clusters

Computing distance between clusters: Complete Link

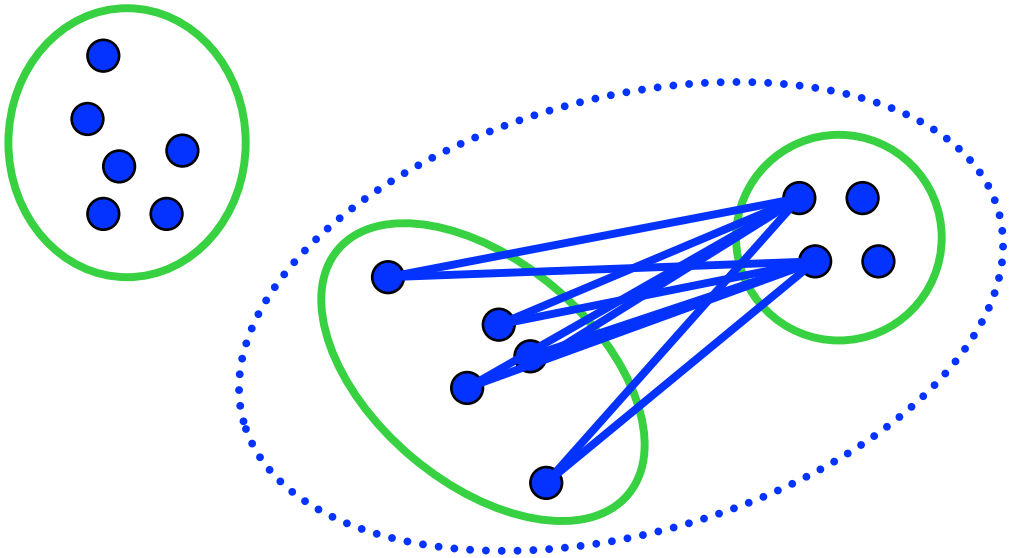
- cluster distance = distance of two farthest members



+ tight clusters

Computing distance between clusters: Average Link

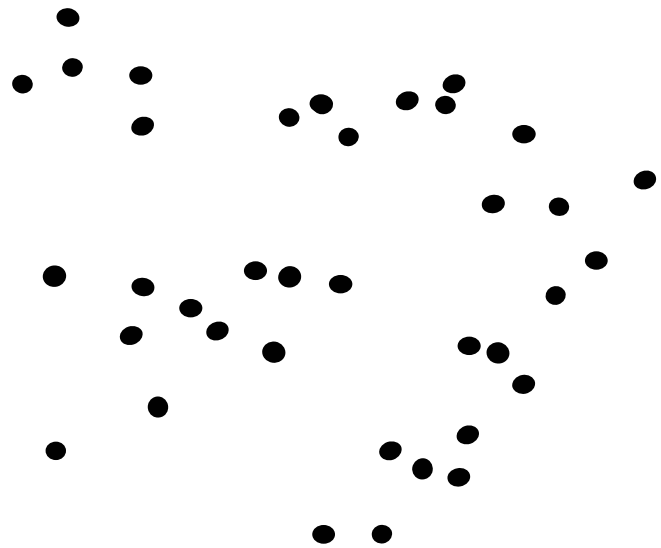
- cluster distance = average distance of all pairs



the most widely used measure

Robust against noise

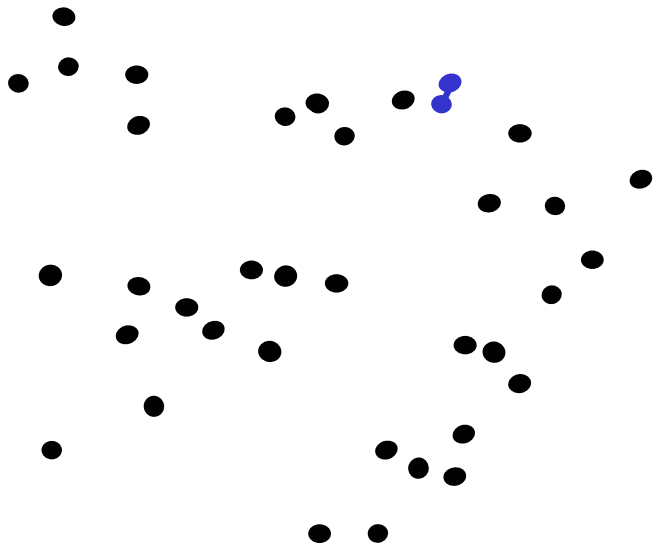
Single Linkage Hierarchical Clustering



1. Say “Every point is its own cluster”

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Single Linkage Hierarchical Clustering

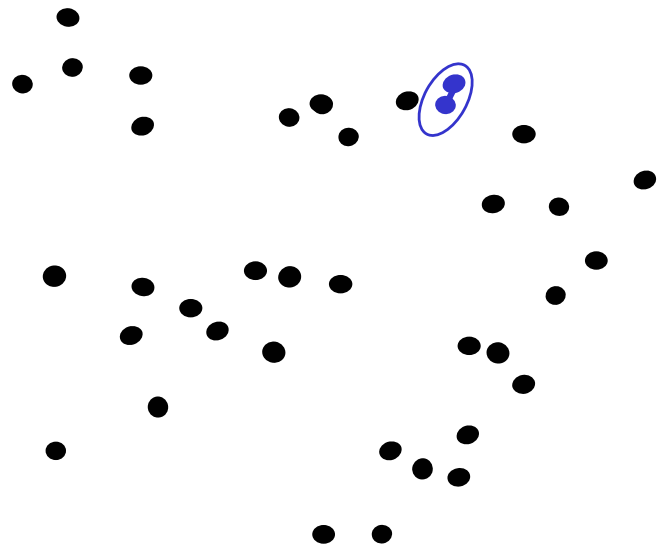


1. Say “Every point is its own cluster”
2. Find “most similar” pair of clusters



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Single Linkage Hierarchical Clustering

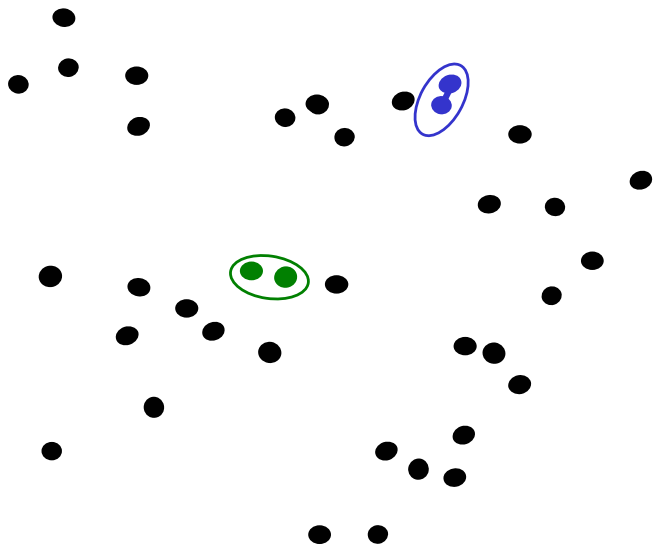


1. Say “Every point is its own cluster”
2. Find “most similar” pair of clusters
3. Merge it into a parent cluster



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Single Linkage Hierarchical Clustering

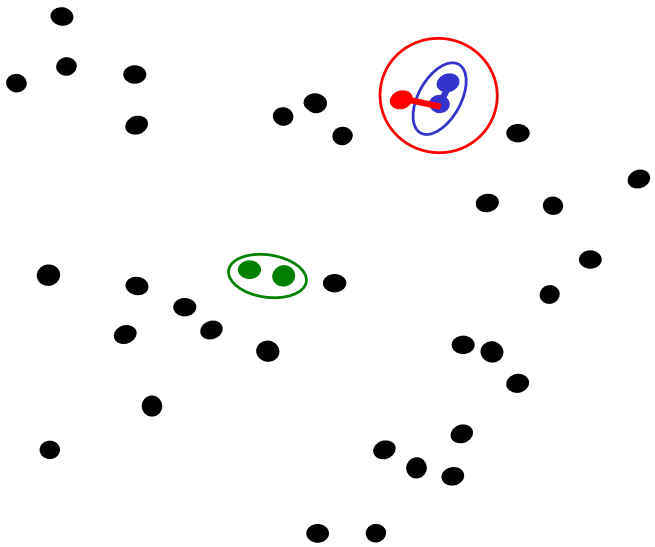


1. Say “Every point is its own cluster”
2. Find “most similar” pair of clusters
3. Merge it into a parent cluster
4. Repeat

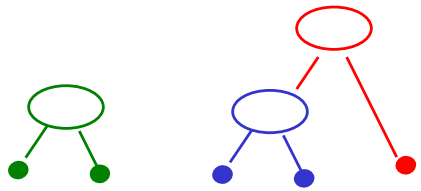


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Single Linkage Hierarchical Clustering

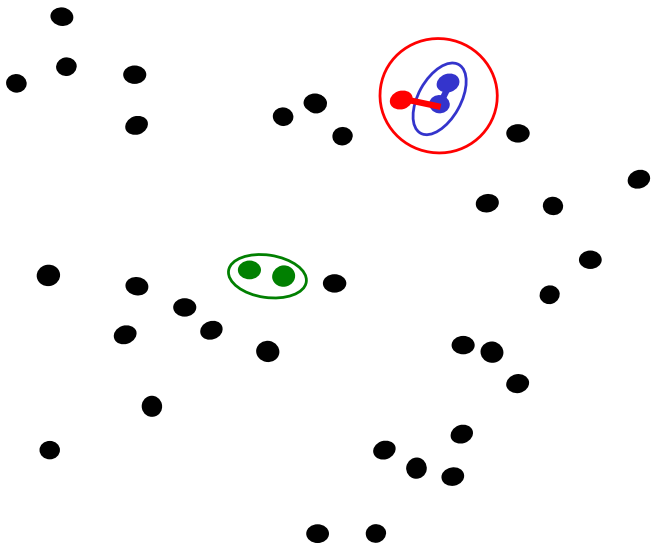


1. Say “Every point is its own cluster”
2. Find “most similar” pair of clusters
3. Merge it into a parent cluster
4. Repeat



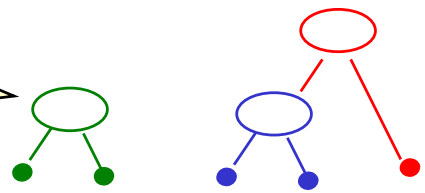
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Single Linkage Hierarchical Clustering



1. Say “Every point is its own cluster”
2. Find “most similar” pair of clusters
3. Merge it into a parent cluster
4. Repeat...until you’ve merged the whole dataset into one cluster

You’re left with a nice dendrogram, or taxonomy, or hierarchy of datapoints (not shown here)



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Also known in the trade as Hierarchical Agglomerative Clustering (note the acronym)

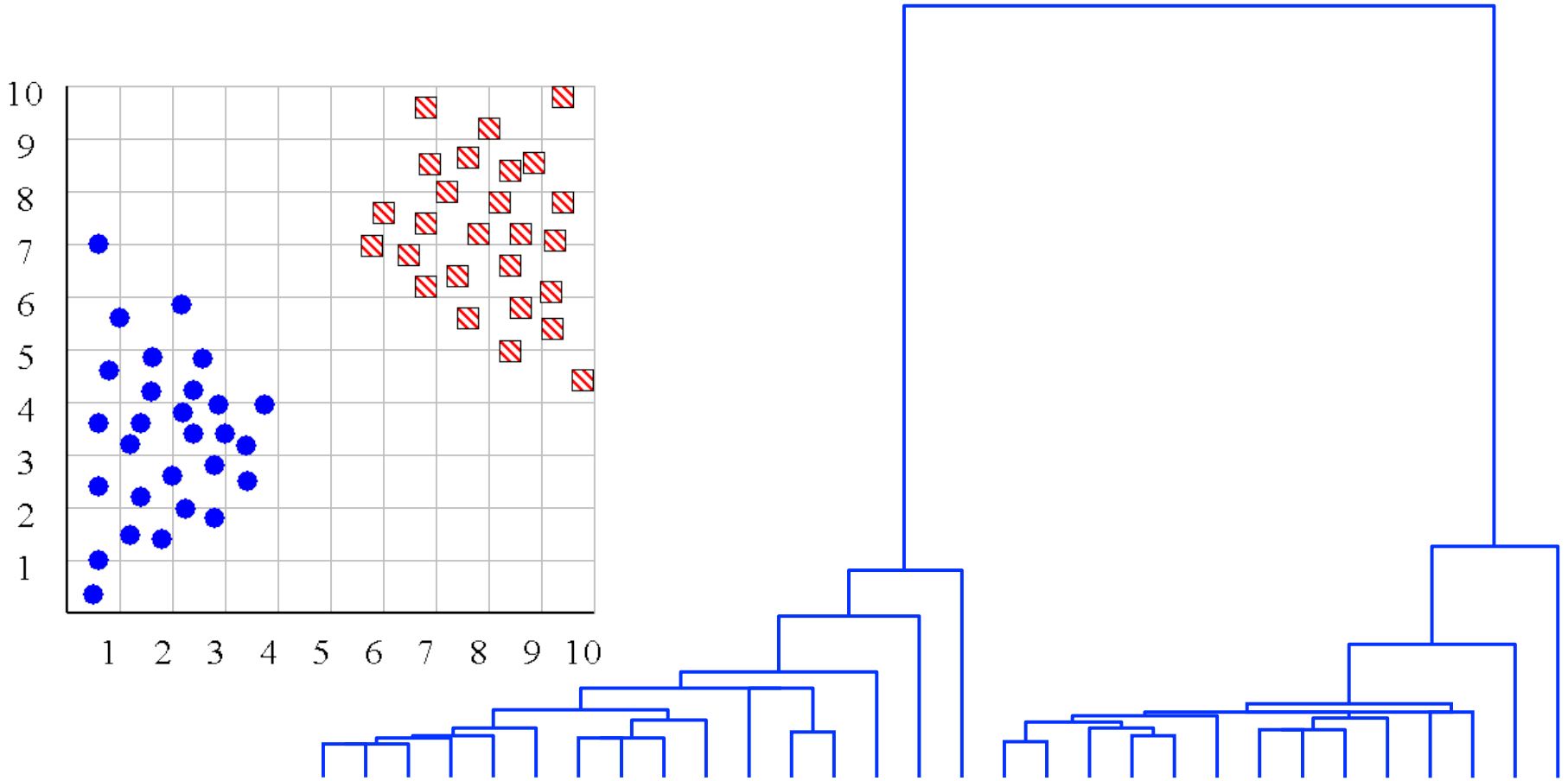
Single Linkage Comments

- It's nice that you get a hierarchy instead of an amorphous collection of groups
- If you want k groups, just cut the (k-1) longest links
- There's no real statistical or information-theoretic foundation to this.

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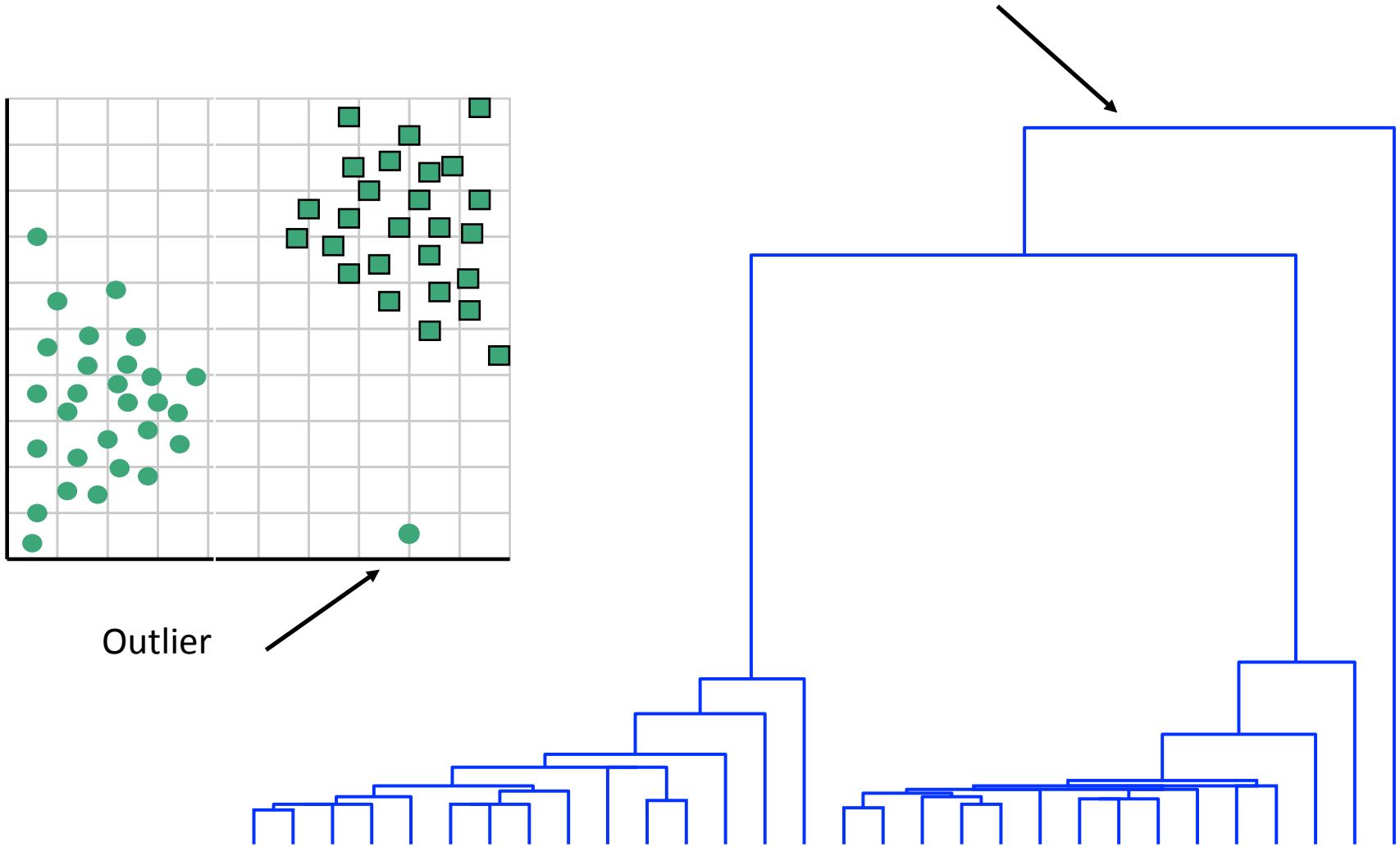
But what are the clusters?

In some cases we can determine the “correct” number of clusters. However, things are rarely this clear cut, unfortunately.



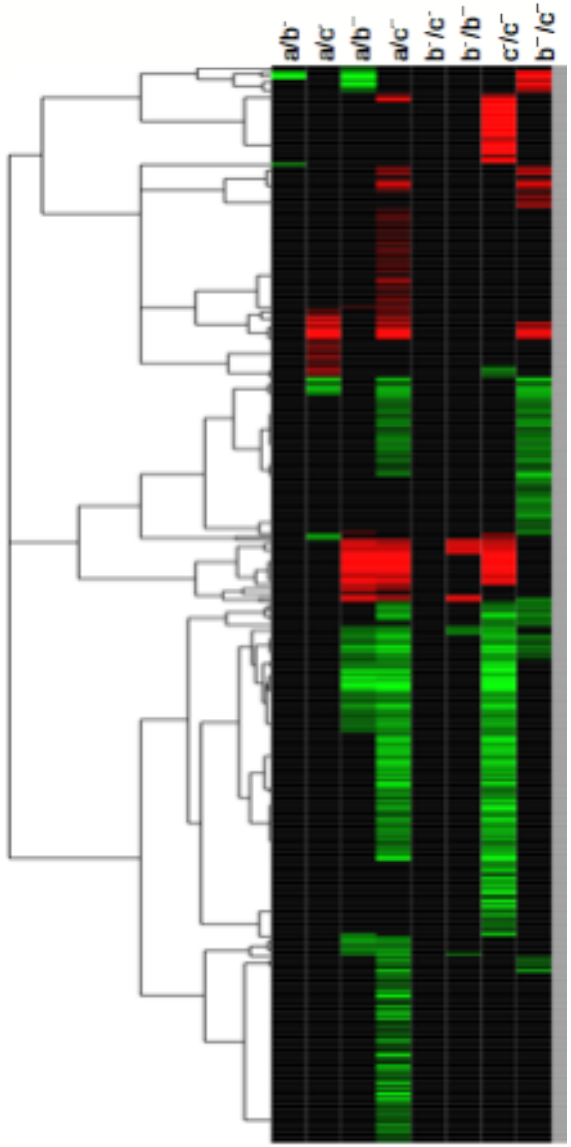
One potential use of a dendrogram is to detect outliers

The single isolated branch is suggestive of a data point that is very different to all others



Example: clustering genes

- Microarrays measure the activities of all genes in different conditions
- Group genes that perform the same function
- Clustering genes can help determine new functions for unknown genes
- An early “killer application” in this area
  - The most cited (>12,000) paper in PNAS!



Summary of Hierarchal Clustering Methods

- No need to specify the number of clusters in advance.
- Hierarchical structure maps nicely onto human intuition for some domains
- They do not scale well: time complexity of at least $O(n^2)$, where n is the number of total objects.
- Like any heuristic search algorithms, local optima are a problem.
- Interpretation of results is (very) subjective.

What you should know

- All the details of K-means
- The theory behind K-means as an optimization algorithm
- How K-means can get stuck
- The outline of Hierarchical clustering
- Be able to contrast between which problems would be relatively well/poorly suited to K-means vs Hierarchical clustering

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